

# Bando et al\_Altair Final Clinical Analysis

[Code ▼](#)

```
library(swimplot) library(coxphf) library(grid) library(gtable) library(readr) library(mosaic) library(dplyr) library(survival)
library(survminer) library(gridtext) library(ggplot2) library(scales) library(officer) library(ggthemes) library(tidyverse)
library(gtsummary) library(flextable) library(parameters) library(car) library(grid) library(FSA)
library(ComplexHeatmap) library(readxl) library(janitor) library(rms) library(pROC) library(DT)
```

```
#Demographics Table by Altair Arm
```

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```

rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")

circ_data_subset1 <- circ_data %>%
  select(
    Age.Group,
    Sex,
    PrimSitev2,
    StageA.alt,
    p_hadNeo,
    p_TxAdjAltair,
    ctDNA1mo,
    p_AltBaselineWin,
    BRAF.V600E,
    RAS,
    MSI) %>%
  mutate(
    Age.Group = factor(Age.Group, levels = c("1", "2"), labels = c("<70", ">70")),
    Sex = factor(Sex, levels = c("Male", "Female")),
    PrimSitev2 = factor(PrimSitev2, levels = c("Right-sided colon", "Left-sided colon",
"Rectum")),
    StageA.alt = factor(StageA.alt, levels = c("I", "II", "III", "IV")),
    p_hadNeo = factor(p_hadNeo, levels=c("FALSE","TRUE"), labels = c("No", "Yes")),
    p_TxAdjAltair = factor(p_TxAdjAltair, levels=c("FALSE","TRUE"), labels = c("No", "Yes")),
    ctDNA1mo = factor(ctDNA1mo, levels = c("NEGATIVE", "POSITIVE"), labels = c("Negative", "Positive")),
    p_AltBaselineWin = factor(p_AltBaselineWin, levels = c("MRD", "OnTreatment", "Surveillance")),
    BRAF.V600E = factor(BRAF.V600E, levels = c("WT", "MUT"), labels = c("BRAF wt", "BRAF V600E")),
    RAS = factor(RAS, levels = c("WT", "MUT"), labels = c("RAS wt", "RAS mut")),
    MSI = factor(MSI, levels = c("MSS", "MSI-High")))

circ_data1 <- read.csv("Altair 20240729 Dataset.csv")

circ_data_subset2 <- circ_data1 %>%
  select(
    Age.Group,
    Sex,
    PrimSitev2,
    StageA.alt,
    p_hadNeo,
    p_TxAdjAltair,
    ctDNA1mo,
    p_AltBaselineWin,
    BRAF.V600E,
    RAS,
    MSI,
    altair.Arm) %>%
  mutate(

```

```
Age.Group = factor(Age.Group, levels = c("1", "2"), labels = c("<70", ">70")),
Sex = factor(Sex, levels = c("Male", "Female")),
PrimSitev2 = factor(PrimSitev2, levels = c("Right-sided colon", "Left-sided colon",
"Rectum")),
StageA.alt = factor(StageA.alt, levels = c("I", "II", "III", "IV")),
p_hadNeo = factor(p_hadNeo, levels=c("FALSE","TRUE"), labels = c("No", "Yes")),
p_TxAdjAltair = factor(p_TxAdjAltair, levels=c("FALSE","TRUE"), labels = c("No", "Yes")),
ctDNA1mo = factor(ctDNA1mo, levels = c("NEGATIVE", "POSITIVE"), labels = c("Negative", "Positive")),
p_AltBaselineWin = factor(p_AltBaselineWin, levels = c("MRD", "OnTreatment", "Surveillance")),
BRAF.V600E = factor(BRAF.V600E, levels = c("WT", "MUT"), labels = c("BRAF wt", "BRAF V600E")),
RAS = factor(RAS, levels = c("WT", "MUT"), labels = c("RAS wt", "RAS mut")),
MSI = factor(MSI, levels = c("MSS", "MSI-High")),
altair.Arm = factor(altair.Arm, levels=c("Control","Experimental"), labels = c("Placebo", "FTD/TPI"))
Overall <- circ_data_subset1 %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{median} ({min} - {max})",
      all_categorical() ~ "{n} ({p}%)" ) %>%
    bold_labels()
  )
Overall
```

| Characteristic     | N = 243 <sup>1</sup> |
|--------------------|----------------------|
| Age.Group          |                      |
| <70                | 155 (64%)            |
| >70                | 88 (36%)             |
| Sex                |                      |
| Male               | 142 (58%)            |
| Female             | 101 (42%)            |
| PrimSitev2         |                      |
| Right-sided colon  | 60 (26%)             |
| Left-sided colon   | 127 (55%)            |
| Rectum             | 46 (20%)             |
| Unknown            | 10                   |
| StageA.alt         |                      |
| <sup>1</sup> n (%) |                      |

| Characteristic     | N = 243 <sup>1</sup> |
|--------------------|----------------------|
| I                  | 10 (4.1%)            |
| II                 | 58 (24%)             |
| III                | 109 (45%)            |
| IV                 | 66 (27%)             |
| p_hadNeo           | 87 (36%)             |
| p_TxAdjAltair      | 112 (46%)            |
| ctDNA1mo           |                      |
| Negative           | 113 (47%)            |
| Positive           | 130 (53%)            |
| p_AltBaselineWin   |                      |
| MRD                | 58 (24%)             |
| OnTreatment        | 33 (14%)             |
| Surveillance       | 152 (63%)            |
| BRAF.V600E         |                      |
| BRAF wt            | 234 (96%)            |
| BRAF V600E         | 9 (3.7%)             |
| RAS                |                      |
| RAS wt             | 148 (61%)            |
| RAS mut            | 95 (39%)             |
| MSI                |                      |
| MSS                | 238 (98%)            |
| MSI-High           | 5 (2.1%)             |
| <sup>1</sup> n (%) |                      |

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```

ByctDNA_MRD <- circ_data_subset2 %>%
  tbl_summary(
    by = altair.Arm, # add this line to subgroup by altair.Arm
    statistic = list(
      all_continuous() ~ "{median} ({min} - {max})",
      all_categorical() ~ "{n} ({p}%)" ) %>%
    add_p() %>%
    bold_labels()
ByctDNA_MRD

```

| Characteristic       | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> | p-value <sup>2</sup> |
|----------------------|---------------------------------|---------------------------------|----------------------|
| <b>Age.Group</b>     |                                 |                                 | 0.8                  |
| <70                  | 78 (64%)                        | 77 (63%)                        |                      |
| >70                  | 43 (36%)                        | 45 (37%)                        |                      |
| <b>Sex</b>           |                                 |                                 | >0.9                 |
| Male                 | 71 (59%)                        | 71 (58%)                        |                      |
| Female               | 50 (41%)                        | 51 (42%)                        |                      |
| <b>PrimSitev2</b>    |                                 |                                 | >0.9                 |
| Right-sided colon    | 29 (25%)                        | 31 (26%)                        |                      |
| Left-sided colon     | 64 (56%)                        | 63 (53%)                        |                      |
| Rectum               | 22 (19%)                        | 24 (20%)                        |                      |
| Unknown              | 6                               | 4                               |                      |
| <b>StageA.alt</b>    |                                 |                                 | 0.6                  |
| I                    | 3 (2.5%)                        | 7 (5.7%)                        |                      |
| II                   | 30 (25%)                        | 28 (23%)                        |                      |
| III                  | 56 (46%)                        | 53 (43%)                        |                      |
| IV                   | 32 (26%)                        | 34 (28%)                        |                      |
| <b>p_hadNeo</b>      | 41 (34%)                        | 46 (38%)                        | 0.5                  |
| <b>p_TxAdjAltair</b> | 56 (46%)                        | 56 (46%)                        | >0.9                 |

<sup>1</sup> n (%)

<sup>2</sup> Pearson's Chi-squared test; Fisher's exact test

| Characteristic                                               | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> | p-value <sup>2</sup> |
|--------------------------------------------------------------|---------------------------------|---------------------------------|----------------------|
| ctDNA1mo                                                     |                                 |                                 | >0.9                 |
| Negative                                                     | 56 (46%)                        | 57 (47%)                        |                      |
| Positive                                                     | 65 (54%)                        | 65 (53%)                        |                      |
| p_AltBaselineWin                                             |                                 |                                 | 0.5                  |
| MRD                                                          | 30 (25%)                        | 28 (23%)                        |                      |
| OnTreatment                                                  | 19 (16%)                        | 14 (11%)                        |                      |
| Surveillance                                                 | 72 (60%)                        | 80 (66%)                        |                      |
| BRAF.V600E                                                   |                                 |                                 | 0.5                  |
| BRAF wt                                                      | 118 (98%)                       | 116 (95%)                       |                      |
| BRAF V600E                                                   | 3 (2.5%)                        | 6 (4.9%)                        |                      |
| RAS                                                          |                                 |                                 | 0.2                  |
| RAS wt                                                       | 79 (65%)                        | 69 (57%)                        |                      |
| RAS mut                                                      | 42 (35%)                        | 53 (43%)                        |                      |
| MSI                                                          |                                 |                                 | 0.7                  |
| MSS                                                          | 118 (98%)                       | 120 (98%)                       |                      |
| MSI-High                                                     | 3 (2.5%)                        | 2 (1.6%)                        |                      |
| <sup>1</sup> n (%)                                           |                                 |                                 |                      |
| <sup>2</sup> Pearson's Chi-squared test; Fisher's exact test |                                 |                                 |                      |

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```
merged_table <- tbl_merge(tbls=list(Overall, ByctDNA_MRD))
merged_table
```

|                                                              | Table 1              | Table 2                         |                                                      |
|--------------------------------------------------------------|----------------------|---------------------------------|------------------------------------------------------|
| Characteristic                                               | N = 243 <sup>1</sup> | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> p-value <sup>2</sup> |
| Age.Group                                                    |                      |                                 | 0.8                                                  |
| <sup>1</sup> n (%)                                           |                      |                                 |                                                      |
| <sup>2</sup> Pearson's Chi-squared test; Fisher's exact test |                      |                                 |                                                      |

| Characteristic                                               | Table 1              | Table 2                         |                                 | p-value <sup>2</sup> |
|--------------------------------------------------------------|----------------------|---------------------------------|---------------------------------|----------------------|
|                                                              | N = 243 <sup>1</sup> | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> |                      |
| <70                                                          | 155 (64%)            | 78 (64%)                        | 77 (63%)                        |                      |
| >70                                                          | 88 (36%)             | 43 (36%)                        | 45 (37%)                        |                      |
| <b>Sex</b>                                                   |                      |                                 |                                 | >0.9                 |
| Male                                                         | 142 (58%)            | 71 (59%)                        | 71 (58%)                        |                      |
| Female                                                       | 101 (42%)            | 50 (41%)                        | 51 (42%)                        |                      |
| <b>PrimSitev2</b>                                            |                      |                                 |                                 | >0.9                 |
| Right-sided colon                                            | 60 (26%)             | 29 (25%)                        | 31 (26%)                        |                      |
| Left-sided colon                                             | 127 (55%)            | 64 (56%)                        | 63 (53%)                        |                      |
| Rectum                                                       | 46 (20%)             | 22 (19%)                        | 24 (20%)                        |                      |
| Unknown                                                      | 10                   | 6                               | 4                               |                      |
| <b>StageA.alt</b>                                            |                      |                                 |                                 | 0.6                  |
| I                                                            | 10 (4.1%)            | 3 (2.5%)                        | 7 (5.7%)                        |                      |
| II                                                           | 58 (24%)             | 30 (25%)                        | 28 (23%)                        |                      |
| III                                                          | 109 (45%)            | 56 (46%)                        | 53 (43%)                        |                      |
| IV                                                           | 66 (27%)             | 32 (26%)                        | 34 (28%)                        |                      |
| <b>p_hadNeo</b>                                              | 87 (36%)             | 41 (34%)                        | 46 (38%)                        | 0.5                  |
| <b>p_TxAdjAltair</b>                                         | 112 (46%)            | 56 (46%)                        | 56 (46%)                        | >0.9                 |
| <b>ctDNA1mo</b>                                              |                      |                                 |                                 | >0.9                 |
| Negative                                                     | 113 (47%)            | 56 (46%)                        | 57 (47%)                        |                      |
| Positive                                                     | 130 (53%)            | 65 (54%)                        | 65 (53%)                        |                      |
| <b>p_AltBaselineWin</b>                                      |                      |                                 |                                 | 0.5                  |
| MRD                                                          | 58 (24%)             | 30 (25%)                        | 28 (23%)                        |                      |
| OnTreatment                                                  | 33 (14%)             | 19 (16%)                        | 14 (11%)                        |                      |
| Surveillance                                                 | 152 (63%)            | 72 (60%)                        | 80 (66%)                        |                      |
| <sup>1</sup> n (%)                                           |                      |                                 |                                 |                      |
| <sup>2</sup> Pearson's Chi-squared test; Fisher's exact test |                      |                                 |                                 |                      |

|                                                              | Table 1              | Table 2                         |                                 |                      |
|--------------------------------------------------------------|----------------------|---------------------------------|---------------------------------|----------------------|
| Characteristic                                               | N = 243 <sup>1</sup> | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> | p-value <sup>2</sup> |
| <b>BRAF.V600E</b>                                            |                      |                                 |                                 | 0.5                  |
| BRAF wt                                                      | 234 (96%)            | 118 (98%)                       | 116 (95%)                       |                      |
| BRAF V600E                                                   | 9 (3.7%)             | 3 (2.5%)                        | 6 (4.9%)                        |                      |
| <b>RAS</b>                                                   |                      |                                 |                                 | 0.2                  |
| RAS wt                                                       | 148 (61%)            | 79 (65%)                        | 69 (57%)                        |                      |
| RAS mut                                                      | 95 (39%)             | 42 (35%)                        | 53 (43%)                        |                      |
| <b>MSI</b>                                                   |                      |                                 |                                 | 0.7                  |
| MSS                                                          | 238 (98%)            | 118 (98%)                       | 120 (98%)                       |                      |
| MSI-High                                                     | 5 (2.1%)             | 3 (2.5%)                        | 2 (1.6%)                        |                      |
| <sup>1</sup> n (%)                                           |                      |                                 |                                 |                      |
| <sup>2</sup> Pearson's Chi-squared test; Fisher's exact test |                      |                                 |                                 |                      |

Hide

```
fit1 <- as_flex_table(  
  merged_table,  
  include = everything(),  
  return_calls = FALSE  
)  
fit1
```

|                   | Table 1              | Table 2                         |                                 |                      |
|-------------------|----------------------|---------------------------------|---------------------------------|----------------------|
| Characteristic    | N = 243 <sup>1</sup> | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> | p-value <sup>2</sup> |
| <b>Age.Group</b>  |                      |                                 |                                 | 0.8                  |
| <70               | 155 (64%)            | 78 (64%)                        | 77 (63%)                        |                      |
| >70               | 88 (36%)             | 43 (36%)                        | 45 (37%)                        |                      |
| <b>Sex</b>        |                      |                                 |                                 | >0.9                 |
| Male              | 142 (58%)            | 71 (59%)                        | 71 (58%)                        |                      |
| Female            | 101 (42%)            | 50 (41%)                        | 51 (42%)                        |                      |
| <b>PrimSitev2</b> |                      |                                 |                                 | >0.9                 |

<sup>1</sup>n (%)

<sup>2</sup>Pearson's Chi-squared test; Fisher's exact test



|                         | Table 1              |                                 | Table 2                         |                      |
|-------------------------|----------------------|---------------------------------|---------------------------------|----------------------|
| Characteristic          | N = 243 <sup>1</sup> | Placebo<br>N = 121 <sup>1</sup> | FTD/TPI<br>N = 122 <sup>1</sup> | p-value <sup>2</sup> |
| Right-sided colon       | 60 (26%)             | 29 (25%)                        | 31 (26%)                        |                      |
| Left-sided colon        | 127 (55%)            | 64 (56%)                        | 63 (53%)                        |                      |
| Rectum                  | 46 (20%)             | 22 (19%)                        | 24 (20%)                        |                      |
| Unknown                 | 10                   | 6                               | 4                               |                      |
| <b>StageA.alt</b>       |                      |                                 |                                 | 0.6                  |
| I                       | 10 (4.1%)            | 3 (2.5%)                        | 7 (5.7%)                        |                      |
| II                      | 58 (24%)             | 30 (25%)                        | 28 (23%)                        |                      |
| III                     | 109 (45%)            | 56 (46%)                        | 53 (43%)                        |                      |
| IV                      | 66 (27%)             | 32 (26%)                        | 34 (28%)                        |                      |
| <b>p_hadNeo</b>         | 87 (36%)             | 41 (34%)                        | 46 (38%)                        | 0.5                  |
| <b>p_TxAdjAltair</b>    | 112 (46%)            | 56 (46%)                        | 56 (46%)                        | >0.9                 |
| <b>ctDNA1mo</b>         |                      |                                 |                                 | >0.9                 |
| Negative                | 113 (47%)            | 56 (46%)                        | 57 (47%)                        |                      |
| Positive                | 130 (53%)            | 65 (54%)                        | 65 (53%)                        |                      |
| <b>p_AltBaselineWin</b> |                      |                                 |                                 | 0.5                  |
| MRD                     | 58 (24%)             | 30 (25%)                        | 28 (23%)                        |                      |
| OnTreatment             | 33 (14%)             | 19 (16%)                        | 14 (11%)                        |                      |
| Surveillance            | 152 (63%)            | 72 (60%)                        | 80 (66%)                        |                      |
| <b>BRAF.V600E</b>       |                      |                                 |                                 | 0.5                  |
| BRAF wt                 | 234 (96%)            | 118 (98%)                       | 116 (95%)                       |                      |
| BRAF V600E              | 9 (3.7%)             | 3 (2.5%)                        | 6 (4.9%)                        |                      |
| <b>RAS</b>              |                      |                                 |                                 | 0.2                  |
| RAS wt                  | 148 (61%)            | 79 (65%)                        | 69 (57%)                        |                      |
| RAS mut                 | 95 (39%)             | 42 (35%)                        | 53 (43%)                        |                      |
| <b>MSI</b>              |                      |                                 |                                 | 0.7                  |
| MSS                     | 238 (98%)            | 118 (98%)                       | 120 (98%)                       |                      |
| MSI-High                | 5 (2.1%)             | 3 (2.5%)                        | 2 (1.6%)                        |                      |

<sup>1</sup>n (%)<sup>2</sup>Pearson's Chi-squared test; Fisher's exact test

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```
save_as_docx(fit1, path = "~/Downloads/merged_table.docx")
```

#Median enrollment MTM/mL in the complete cohort

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```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_datadf <- as.data.frame(circ_data)

median_val <- median(circ_data$p_AltBaselineMTM, na.rm = TRUE)
q1_val <- quantile(circ_data$p_AltBaselineMTM, 0.25, na.rm = TRUE)
q3_val <- quantile(circ_data$p_AltBaselineMTM, 0.75, na.rm = TRUE)
range_val <- range(circ_data$p_AltBaselineMTM, na.rm = TRUE)
cat("Median:", format(median_val, digits = 4), "\n")
```

Median: 0.4

[Hide](#)

```
cat("Q1 (25th percentile):", q1_val, "\n")
```

Q1 (25th percentile): 0.1096503

[Hide](#)

```
cat("Q3 (75th percentile):", q3_val, "\n")
```

Q3 (75th percentile): 1.767375

[Hide](#)

```
cat("Range:", range_val, "\n")
```

Range: 0.02627334 250.47

#Median enrollment MTM/mL by Stage

[Hide](#)

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_datadf <- as.data.frame(circ_data)
circ_data$StageA.alt <- factor(circ_data$StageA.alt, levels = c("I", "II", "III", "IV"))

stage_summary <- circ_data %>%
  group_by(StageA.alt) %>%
  summarise(
    Median = median(p_AltBaselineMTM, na.rm = TRUE),
    Q1 = quantile(p_AltBaselineMTM, 0.25, na.rm = TRUE),
    Q3 = quantile(p_AltBaselineMTM, 0.75, na.rm = TRUE),
    Min = min(p_AltBaselineMTM, na.rm = TRUE),
    Max = max(p_AltBaselineMTM, na.rm = TRUE)
  ) %>%
  ungroup()
print(stage_summary)
```

```
# A tibble: 4 × 6
  StageA.alt Median      Q1      Q3      Min      Max
  <fct>      <dbl> <dbl> <dbl> <dbl> <dbl>
1 I          0.190 0.0556 1.29  0.0302 121.
2 II         0.294 0.0879 0.981 0.0263 166.
3 III        0.379 0.118  1.42  0.0290 250.
4 IV         0.680 0.185  2.60  0.0344 40.4
```

Hide

```
# Global test
kruskal_test <- kruskal.test(p_AltBaselineMTM ~ StageA.alt, data = circ_data)
print(kruskal_test)
```

Kruskal-Wallis rank sum test

```
data: p_AltBaselineMTM by StageA.alt
Kruskal-Wallis chi-squared = 6.977, df = 3, p-value = 0.07264
```

Hide

```
# Pairwise comparisons with p-value adjustment (to avoid false positives)
pairwise_results <- dunnTest(p_AltBaselineMTM ~ StageA.alt,
                             data = circ_data,
                             method = "bh") # "bh" is the Benjamini-Hochberg adjustment

print(pairwise_results)
```

|   | Comparison | Z          | P.unadj    | P.adj      |
|---|------------|------------|------------|------------|
| 1 | I - II     | -0.1752904 | 0.86085144 | 0.86085144 |
| 2 | I - III    | -0.7898581 | 0.42961066 | 0.51553279 |
| 3 | II - III   | -1.2364613 | 0.21628714 | 0.32443072 |
| 4 | I - IV     | -1.4917154 | 0.13577377 | 0.27154754 |
| 5 | II - IV    | -2.4790389 | 0.01317369 | 0.07904216 |
| 6 | III - IV   | -1.5722337 | 0.11589635 | 0.34768906 |

#Median enrollment MTM/mL by enrollment window

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```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_datadf <- as.data.frame(circ_data)
circ_data$p_AltBaselineWin <- factor(circ_data$p_AltBaselineWin, levels = c("MRD", "OnTreatment", "Surveillance"))

stage_summary <- circ_data %>%
  group_by(p_AltBaselineWin) %>%
  summarise(
    Median = median(p_AltBaselineMTM, na.rm = TRUE),
    Q1 = quantile(p_AltBaselineMTM, 0.25, na.rm = TRUE),
    Q3 = quantile(p_AltBaselineMTM, 0.75, na.rm = TRUE),
    Min = min(p_AltBaselineMTM, na.rm = TRUE),
    Max = max(p_AltBaselineMTM, na.rm = TRUE)
  ) %>%
  ungroup()
print(stage_summary)
```

```
# A tibble: 3 × 6
  p_AltBaselineWin Median      Q1      Q3      Min      Max
  <fct>             <dbl>   <dbl> <dbl>   <dbl>   <dbl>
1 MRD               0.972 0.208  4.22  0.0265  121.
2 OnTreatment       0.822 0.232  2.12  0.0602  166.
3 Surveillance      0.240 0.0880 0.985 0.0263  250.
```

#Median enrollment MTM/mL by enrollment window in each treatment arm

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```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_datadf <- as.data.frame(circ_data)
circ_data$p_AltBaselineWin <- factor(circ_data$p_AltBaselineWin, levels = c("MRD", "OnTreatment", "Surveillance"))
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels = c("Control", "Experimental"), labels = c("Placebo", "FTD/TPI"))

# 1. Summary statistics by p_AltBaselineWin
stage_summary <- circ_data %>%
  group_by(p_AltBaselineWin) %>%
  summarise(
    Median = median(p_AltBaselineMTM, na.rm = TRUE),
    Q1 = quantile(p_AltBaselineMTM, 0.25, na.rm = TRUE),
    Q3 = quantile(p_AltBaselineMTM, 0.75, na.rm = TRUE),
    Min = min(p_AltBaselineMTM, na.rm = TRUE),
    Max = max(p_AltBaselineMTM, na.rm = TRUE)
  ) %>%
  ungroup()
print("Summary by p_AltBaselineWin:")
```

```
[1] "Summary by p_AltBaselineWin:"
```

[Hide](#)

```
print(stage_summary)
```

```
# A tibble: 3 × 6
  p_AltBaselineWin Median      Q1      Q3      Min      Max
  <fct>             <dbl> <dbl> <dbl> <dbl> <dbl>
1 MRD               0.972 0.208 4.22  0.0265 121.
2 OnTreatment       0.822 0.232 2.12  0.0602 166.
3 Surveillance      0.240 0.0880 0.985 0.0263 250.
```

[Hide](#)

```
# 2. Median by p_AltBaselineWin and altair.Arm
arm_stage_summary <- circ_data %>%
  group_by(p_AltBaselineWin, altair.Arm) %>%
  summarise(
    Median = median(p_AltBaselineMTM, na.rm = TRUE),
    n = n()
  ) %>%
  ungroup()
print("Median p_AltBaselineMTM by p_AltBaselineWin and altair.Arm:")
```

```
[1] "Median p_AltBaselineMTM by p_AltBaselineWin and altair.Arm:"
```

Hide

```
print(arm_stage_summary)
```

```
# A tibble: 6 × 4
  p_AltBaselineWin altair.Arm Median      n
  <fct>            <fct>      <dbl> <int>
1 MRD              Placebo    0.972   30
2 MRD              FTD/TPI    0.875   28
3 OnTreatment      Placebo    0.908   19
4 OnTreatment      FTD/TPI    0.741   14
5 Surveillance     Placebo    0.169   72
6 Surveillance     FTD/TPI    0.293   80
```

Hide

```
# 3. Wilcoxon test p-values for Control vs Experimental within each p_AltBaselineWin
p_values <- circ_data %>%
  filter(!is.na(p_AltBaselineMTM), !is.na(altair.Arm), !is.na(p_AltBaselineWin)) %>%
  group_by(p_AltBaselineWin) %>%
  summarise(
    p_value = tryCatch(
      wilcox.test(p_AltBaselineMTM ~ altair.Arm)$p.value,
      error = function(e) NA
    )
  )
print("Wilcoxon test p-values by p_AltBaselineWin:")
```

```
[1] "Wilcoxon test p-values by p_AltBaselineWin:"
```

Hide

```
print(p_values)
```

```
# A tibble: 3 × 2
  p_AltBaselineWin p_value
  <fct>            <dbl>
1 MRD              0.616
2 OnTreatment      0.529
3 Surveillance     0.220
```

```
#Median enrollment MTM/mL in Stage IV vs Non-Stage IV
```

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[!(circ_data$StageA.alt %in% c("I", "II", "III")),]
circ_datadf <- as.data.frame(circ_data)

median_val <- median(circ_data$p_AltBaselineMTM, na.rm = TRUE)
q1_val <- quantile(circ_data$p_AltBaselineMTM, 0.25, na.rm = TRUE)
q3_val <- quantile(circ_data$p_AltBaselineMTM, 0.75, na.rm = TRUE)
range_val <- range(circ_data$p_AltBaselineMTM, na.rm = TRUE)
cat("Median:", format(median_val, digits = 4), "\n")
```

Median: 0.6802

[Hide](#)

```
cat("Q1 (25th percentile):", q1_val, "\n")
```

Q1 (25th percentile): 0.1848869

[Hide](#)

```
cat("Q3 (75th percentile):", q3_val, "\n")
```

Q3 (75th percentile): 2.59763

[Hide](#)

```
cat("Range:", range_val, "\n")
```

Range: 0.03444433 40.42593

[Hide](#)

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_datadf <- as.data.frame(circ_data)
circ_data$Stage.Final <- factor(circ_data$Stage.Final, levels = c("I-III", "IV"))
median_MTM <- aggregate(p_AltBaselineMTM ~ Stage.Final, data = circ_data, FUN = median)
print(median_MTM)
```

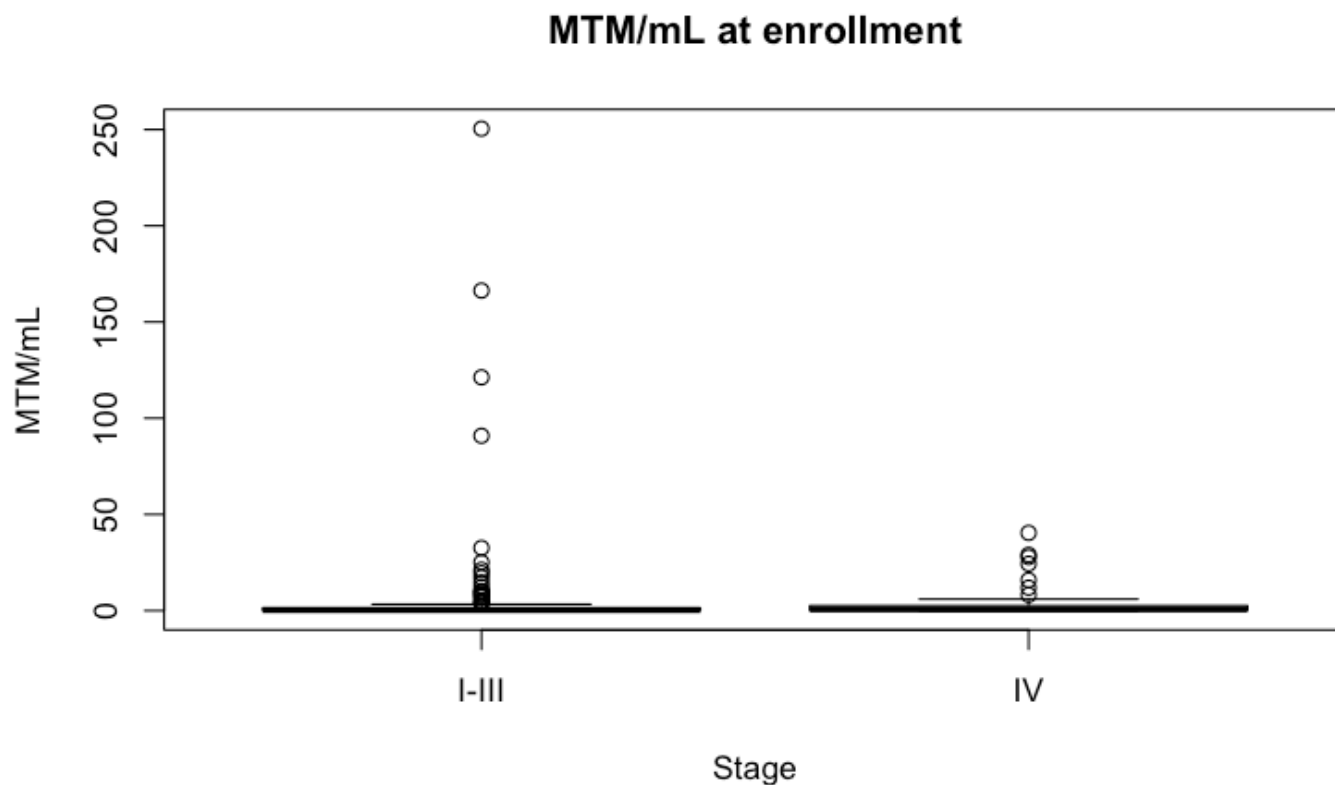
|   | Stage.Final | p_AltBaselineMTM |
|---|-------------|------------------|
| 1 | I-III       | 0.3199376        |
| 2 | IV          | 0.6802357        |

[Hide](#)

```

circ_data$Stage.Final <- factor(circ_data$Stage.Final, levels = c("I-III", "IV"))
boxplot(p_AltBaselineMTM~Stage.Final, data=circ_data, main="MTM/mL at enrollment", xlab
="Stage", ylab="MTM/mL", col="white",border="black")

```


[Hide](#)

```

m1<-wilcox.test(p_AltBaselineMTM ~ Stage.Final, data=circ_data, na.rm=TRUE, exact=FALSE,
conf.int=TRUE)
print(m1)

```

Wilcoxon rank sum test with continuity correction

```

data:  p_AltBaselineMTM by Stage.Final
W = 4740, p-value = 0.02394
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -0.41562942 -0.01469699
sample estimates:
difference in location
      -0.1351274

```

#DFS1 by TAS vs Placebo - All stages & stratified for Stage & ctDNA 1mo post-surgery

[Hide](#)



```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$DFS.months, event = circ\_data\$p\_evtDFS1b) ~ altair.Arm, data = circ\_data)

|                         | n   | events | median | 0.95LCL | 0.95UCL |
|-------------------------|-----|--------|--------|---------|---------|
| altair.Arm=Control      | 121 | 99     | 5.55   | 4.17    | 7.33    |
| altair.Arm=Experimental | 122 | 99     | 9.30   | 7.92    | 10.84   |

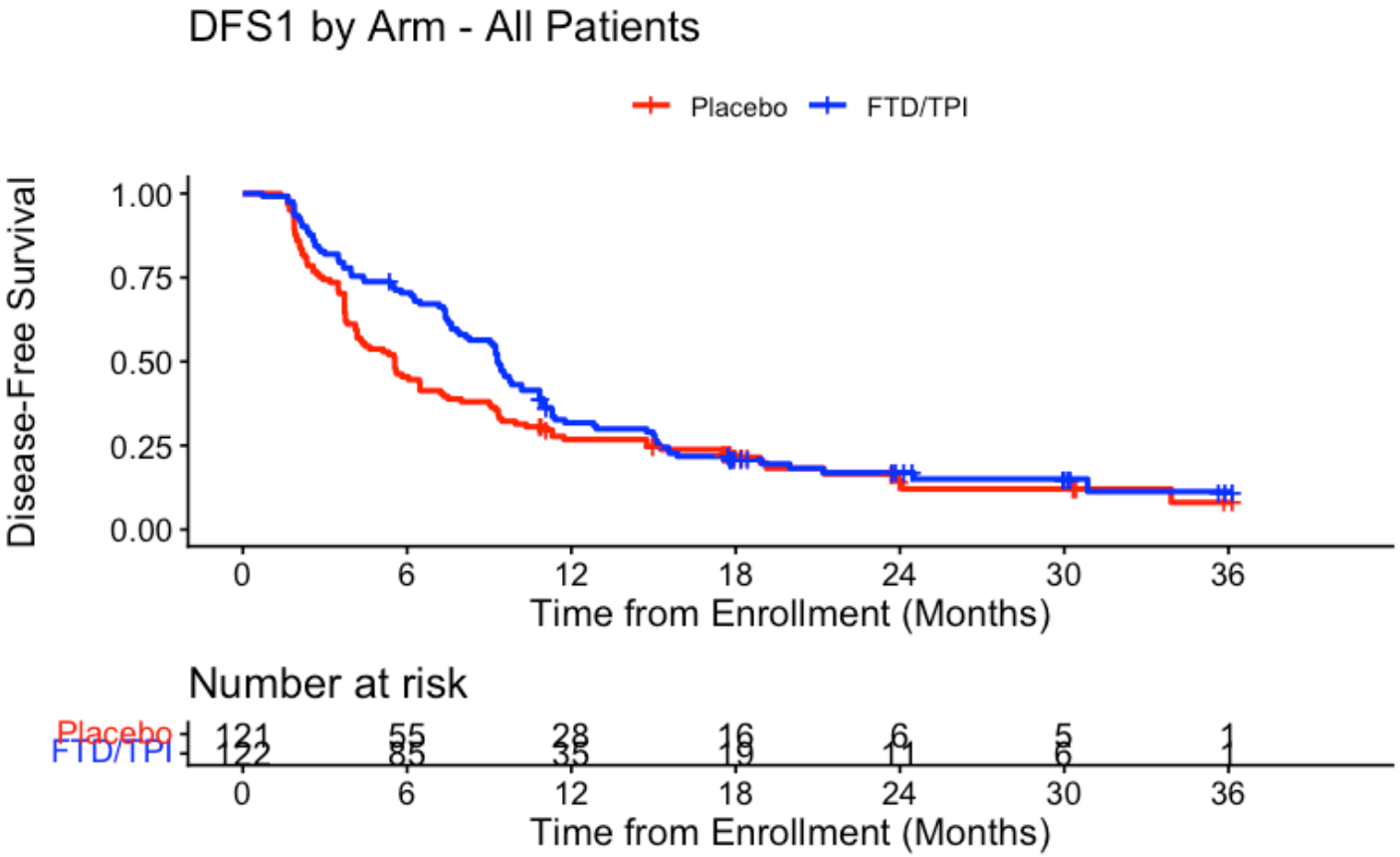
Hide

```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>  <int>   <dbl>      <dbl>
1 Control      121    99    0.818      81.8
2 Experimental 122    99    0.811      81.1
```

Hide

```
surv_object <-Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - All Patients", ylab="Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("Placebo", "FTD/TPI"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

```
      altair.Arm=Control
time  n.risk n.event survival std.err lower 95% CI upper 95% CI
  6      55      66   0.455  0.0453   0.3642   0.540
 12      28      22   0.268  0.0408   0.1916   0.350
 18      16       5   0.215  0.0391   0.1443   0.296
 24       6       4   0.145  0.0394   0.0785   0.231

      altair.Arm=Experimental
time  n.risk n.event survival std.err lower 95% CI upper 95% CI
  6      85      36   0.705  0.0413   0.615    0.777
 12      35      46   0.318  0.0428   0.236    0.402
 18      19      12   0.208  0.0380   0.139    0.287
 24      11       3   0.169  0.0370   0.104    0.248
```

[Hide](#)

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
circ_data$Disease.Stage <- factor(circ_data$Disease.Stage, levels=c("Stage II or lower", "Stage III", "M1"))
circ_data$ctDNA1mo <- factor(circ_data$ctDNA1mo, levels=c("NEGATIVE","POSITIVE"), labels = c("Negative", "Positive"))
cox_fit_stratified <- coxph(surv_object ~ altair.Arm + strata(Disease.Stage) + strata(ctDNA1mo), data = circ_data)
summary(cox_fit_stratified)
```

```
Call:
coxph(formula = surv_object ~ altair.Arm + strata(Disease.Stage) +
  strata(ctDNA1mo), data = circ_data)
```

n= 243, number of events= 198

```
      coef exp(coef) se(coef)      z Pr(>|z|)
altair.ArmFTD/TPI -0.2337   0.7916  0.1448 -1.614   0.107
```

```
      exp(coef) exp(-coef) lower .95 upper .95
altair.ArmFTD/TPI   0.7916    1.263    0.596    1.051
```

Concordance= 0.555 (se = 0.021 )

Likelihood ratio test= 2.6 on 1 df, p=0.1

Wald test = 2.6 on 1 df, p=0.1

Score (logrank) test = 2.61 on 1 df, p=0.1

[Hide](#)

```
# Extract values for HR, 95% CI, and p-value
cox_fit_summary_stratified <- summary(cox_fit_stratified)
HR_stratified <- cox_fit_summary_stratified$coefficients[2]
lower_CI_stratified <- cox_fit_summary_stratified$conf.int[3]
upper_CI_stratified <- cox_fit_summary_stratified$conf.int[4]
p_value_stratified <- cox_fit_summary_stratified$coefficients[5]
label_text_stratified <- paste0("HR = ", round(HR_stratified, 2),
                                " (", round(lower_CI_stratified, 2), "-",
                                round(upper_CI_stratified, 2), "); p = ",
                                round(p_value_stratified, 3))

print(label_text_stratified)
```

```
[1] "HR = 0.79 (0.6-1.05); p = 0.107"
```

#DFS1 by TAS vs Placebo - Stage I-III

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[!(circ_data$StageA.alt %in% c("IV")),]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data
= circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
~
      altair.Arm, data = circ_data)
```

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 89 | 67     | 6.05   | 4.63    | 9.23    |
| altair.Arm=Experimental | 88 | 68     | 9.27   | 7.62    | 10.97   |

Hide

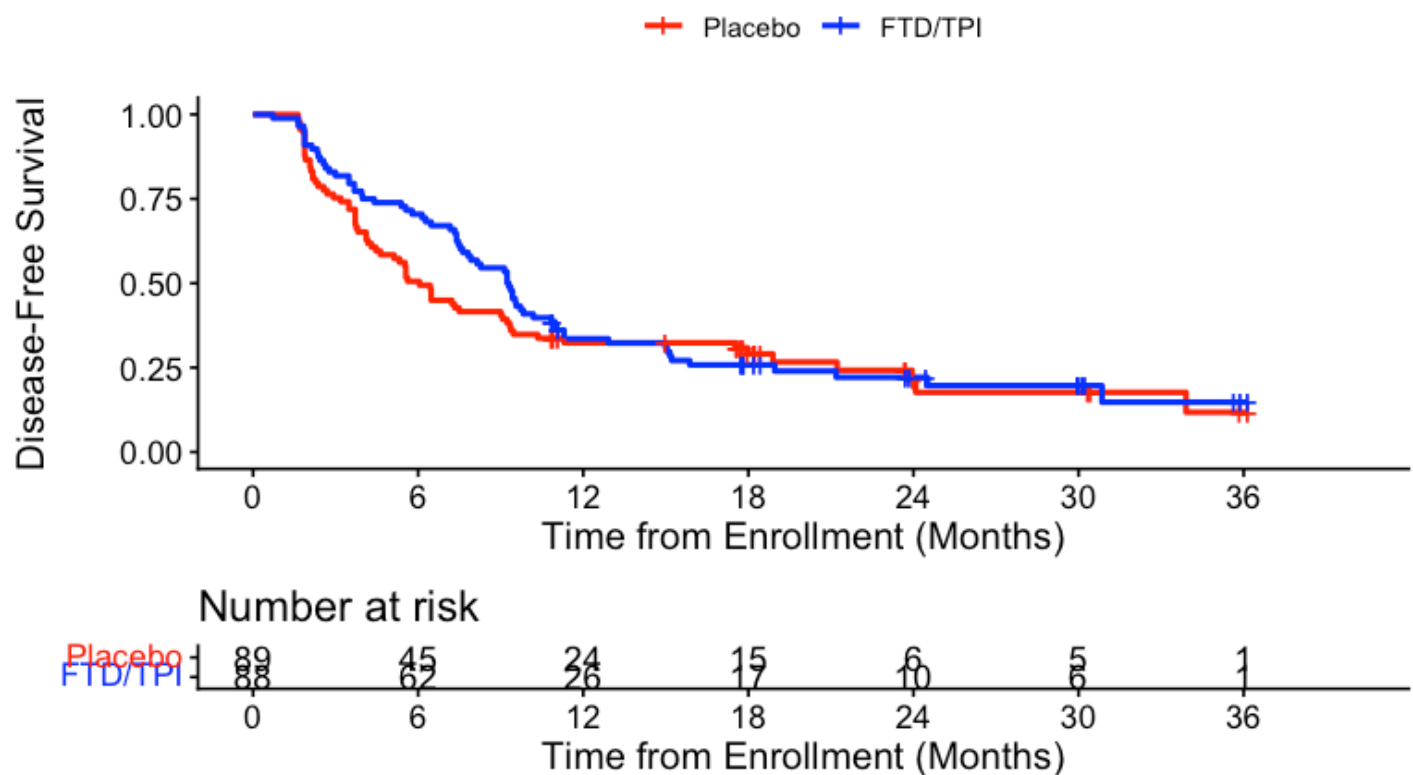
```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm    Total Events Fraction Percentage
  <chr>         <int>   <int>     <dbl>     <dbl>
1 Control         89     67     0.753     75.3
2 Experimental    88     68     0.773     77.3
```

[Hide](#)

```
surv_object <-Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - Stage I-III", ylab= "D
isease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("Placebo", "F
TD/TPI"), legend.title="")
```

## DFS1 by Arm - Stage I-III



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

```
altair.Arm=Control
```

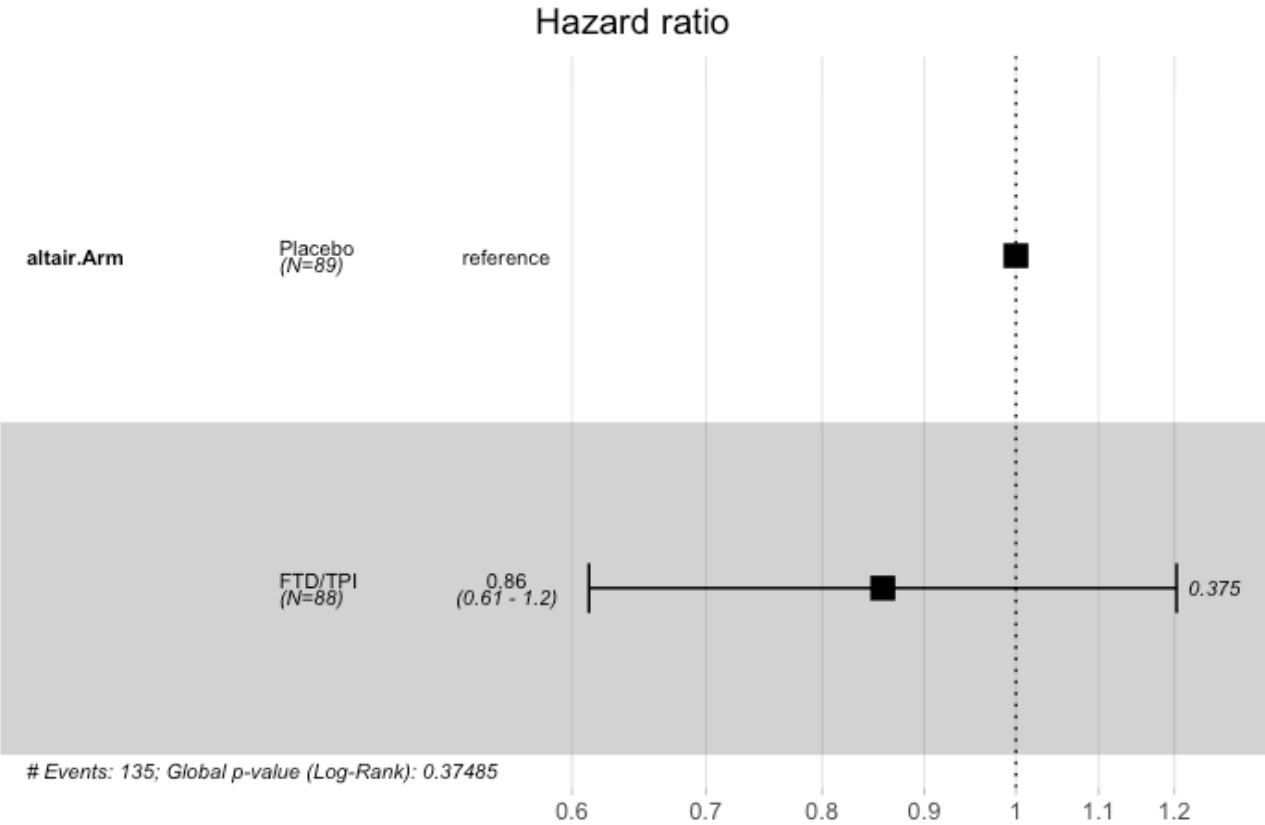
| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 45     | 44      | 0.506    | 0.0530  | 0.398        | 0.604        |
| 12   | 24     | 16      | 0.324    | 0.0499  | 0.229        | 0.422        |
| 18   | 15     | 2       | 0.291    | 0.0500  | 0.197        | 0.391        |
| 24   | 6      | 3       | 0.212    | 0.0537  | 0.118        | 0.324        |

```
altair.Arm=Experimental
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 62     | 26      | 0.705    | 0.0486  | 0.597        | 0.788        |
| 12   | 26     | 32      | 0.336    | 0.0509  | 0.238        | 0.435        |
| 18   | 17     | 6       | 0.258    | 0.0480  | 0.170        | 0.355        |
| 24   | 10     | 2       | 0.221    | 0.0477  | 0.136        | 0.320        |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.Arm, data = circ\_data)

n= 177, number of events= 135

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.1532 | 0.8580    | 0.1725   | -0.888 | 0.375    |

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.858     | 1.166      | 0.6118    | 1.203     |

Concordance= 0.537 (se = 0.023 )  
Likelihood ratio test= 0.79 on 1 df, p=0.4  
Wald test = 0.79 on 1 df, p=0.4  
Score (logrank) test = 0.79 on 1 df, p=0.4

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.86 (0.61-1.2); p = 0.375"
```

#DFS1 by TAS vs Placebo - Stage IV

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[!(circ_data$StageA.alt %in% c("I", "II", "III")),]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$DFS.months, event = circ\_data\$p\_evtDFS1b) ~ altair.Arm, data = circ\_data)

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 32 | 32     | 3.96   | 3.71    | 7.98    |
| altair.Arm=Experimental | 34 | 31     | 9.76   | 7.62    | 11.76   |

Hide

```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

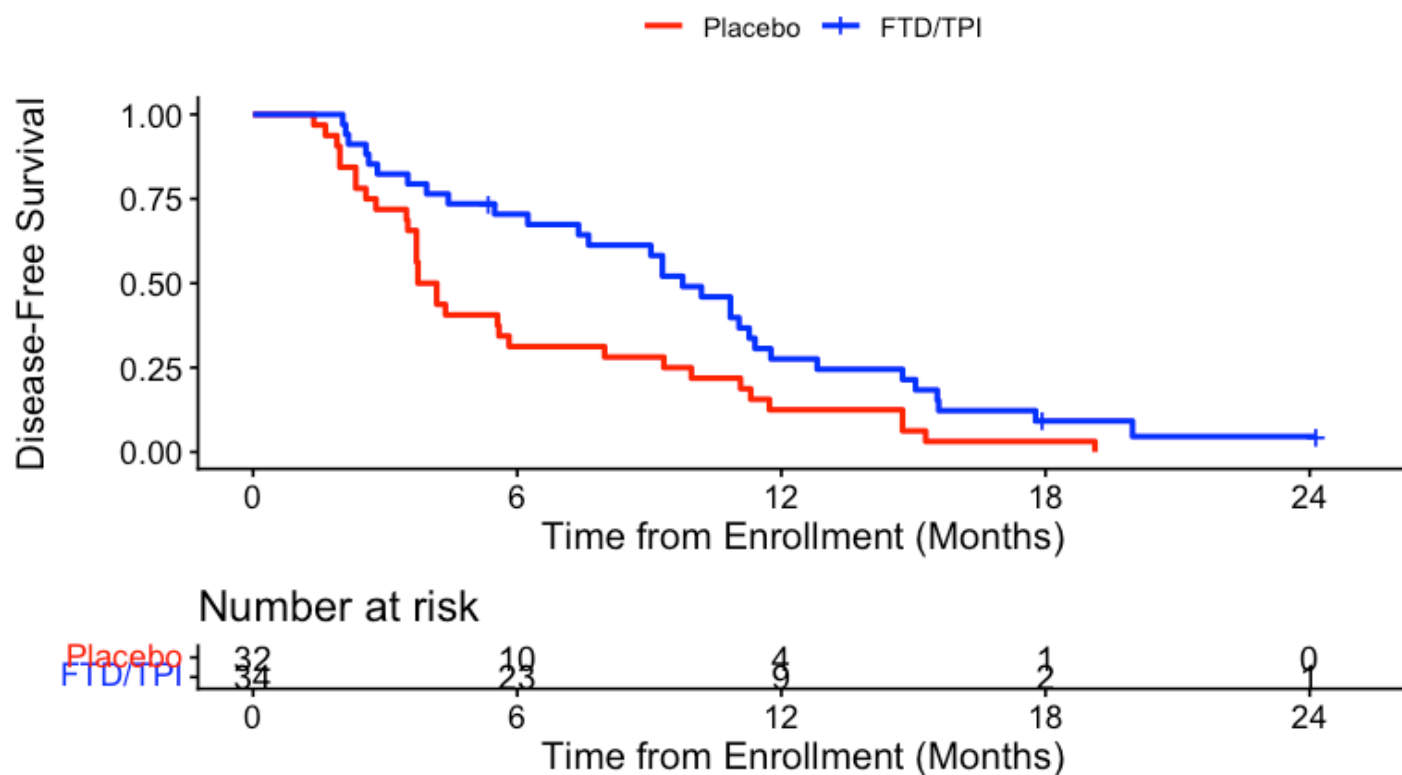


```
# A tibble: 2 × 5
  altair.Arm    Total Events Fraction Percentage
  <chr>         <int>   <int>    <dbl>      <dbl>
1 Control         32     32      1         100
2 Experimental    34     31    0.912     91.2
```

[Hide](#)

```
surv_object <-Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - Stage IV", ylab= "Disease-Free Survival",
xlab="Time from Enrollment (Months)", legend.labs=c("Placebo", "FTD/TPI"), legend.title="")
```

## DFS1 by Arm - Stage IV



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

```
altair.Arm=Control
```

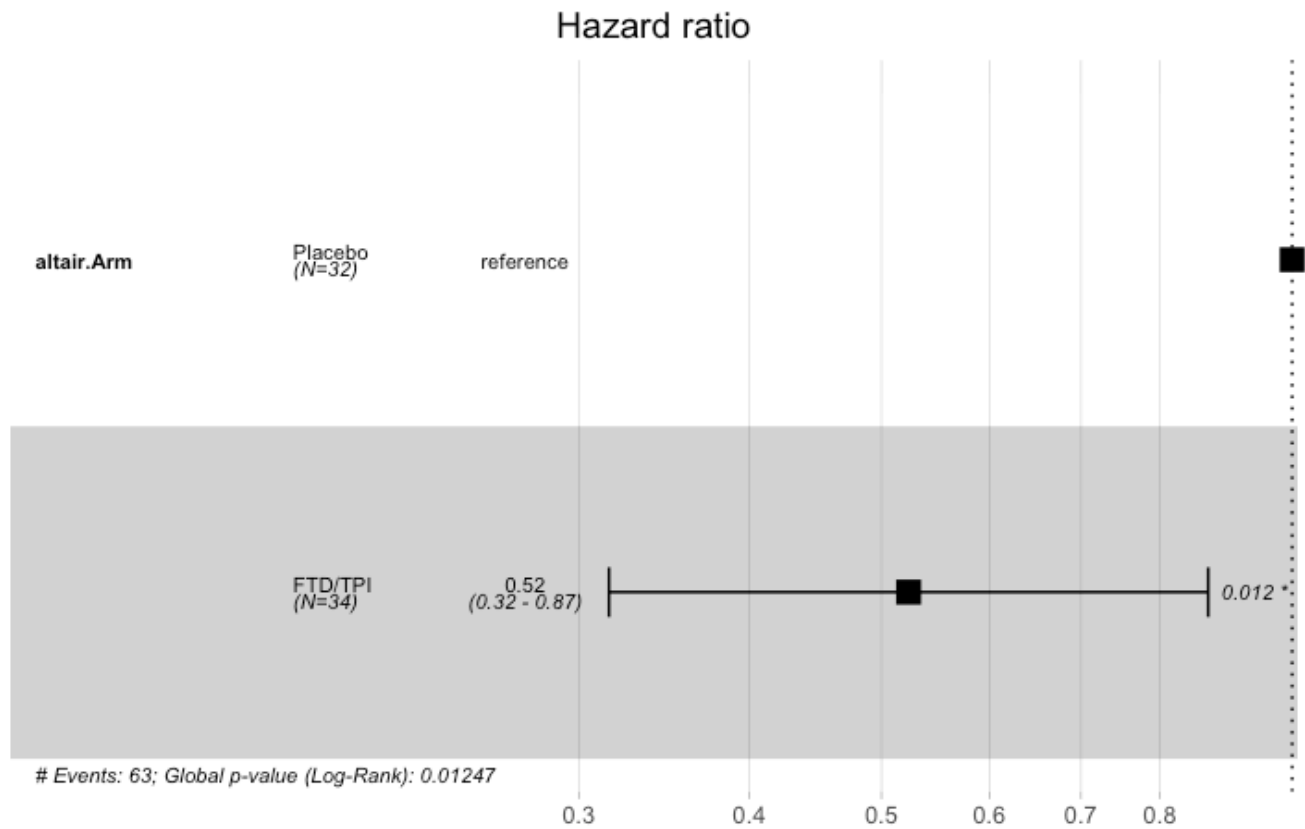
| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 10     | 22      | 0.3125   | 0.0819  | 0.16376      | 0.473        |
| 12   | 4      | 6       | 0.1250   | 0.0585  | 0.03950      | 0.262        |
| 18   | 1      | 3       | 0.0312   | 0.0308  | 0.00237      | 0.137        |

```
altair.Arm=Experimental
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 23     | 10      | 0.7047   | 0.0785  | 0.52049      | 0.829        |
| 12   | 9      | 14      | 0.2757   | 0.0780  | 0.13790      | 0.433        |
| 18   | 2      | 6       | 0.0919   | 0.0505  | 0.02355      | 0.219        |
| 24   | 1      | 1       | 0.0460   | 0.0412  | 0.00431      | 0.175        |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.Arm, data = circ\_data)

n= 66, number of events= 63

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.6474 | 0.5234    | 0.2580   | -2.509 | 0.0121 * |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.5234    | 1.911      | 0.3157    | 0.8679    |

Concordance= 0.596 (se = 0.034 )  
Likelihood ratio test= 6.24 on 1 df, p=0.01  
Wald test = 6.3 on 1 df, p=0.01  
Score (logrank) test = 6.5 on 1 df, p=0.01

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.52 (0.32-0.87); p = 0.012"
```

#DFS1 by TAS vs Placebo - ctDNA positive post-surgery

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$ctDNA1mo == "POSITIVE",]
circ_data <- subset(circ_data, !is.na(p_MRD))
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
~
      altair.Arm, data = circ_data)
```

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 65 | 58     | 4.17   | 3.71    | 6.47    |
| altair.Arm=Experimental | 65 | 55     | 7.92   | 7.16    | 10.84   |

Hide

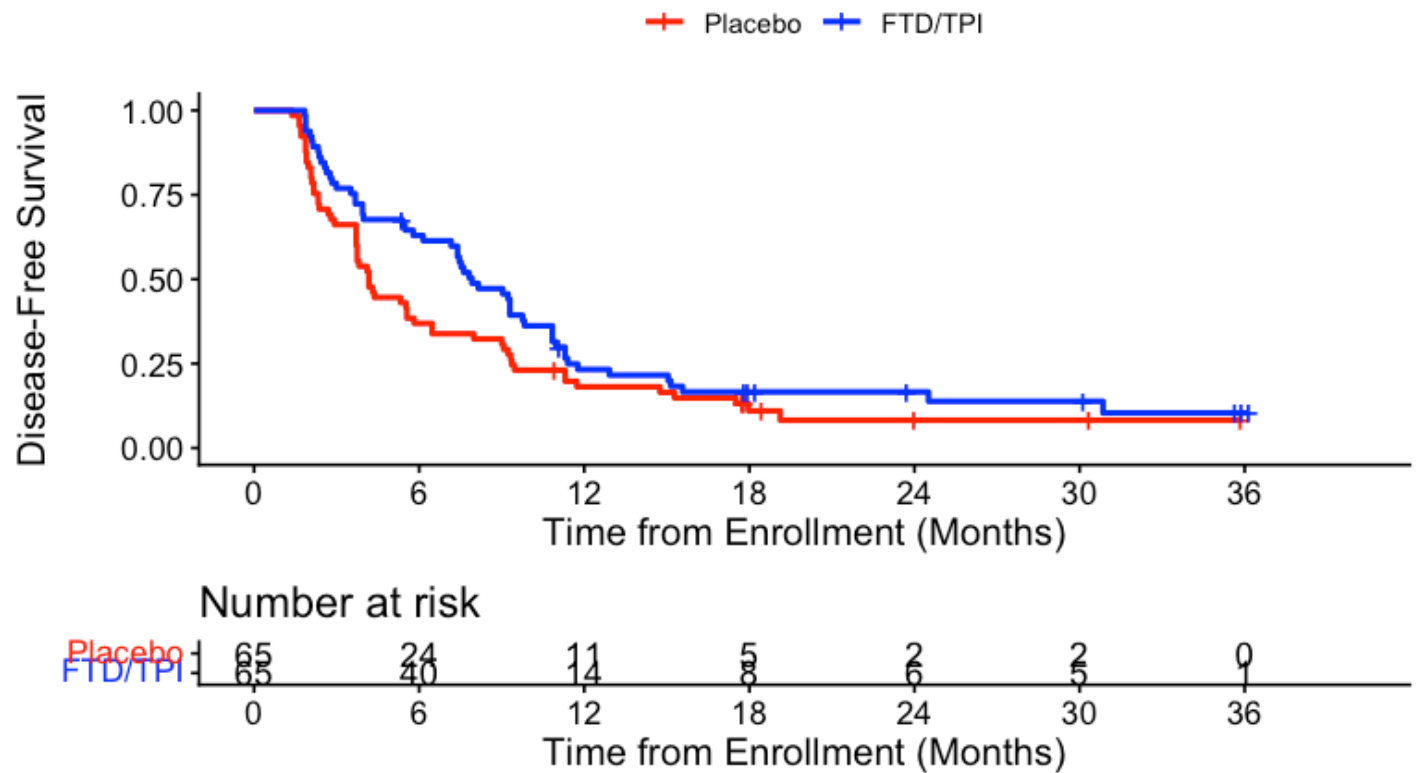
```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>   <int>   <dbl>     <dbl>
1 Control         65     58    0.892     89.2
2 Experimental    65     55    0.846     84.6
```

[Hide](#)

```
surv_object <-Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - ctDNA positive post-surgery",
ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs
=c("Placebo", "FTD/TPI"), legend.title="")
```

DFS1 by Arm - ctDNA positive post-surgery



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

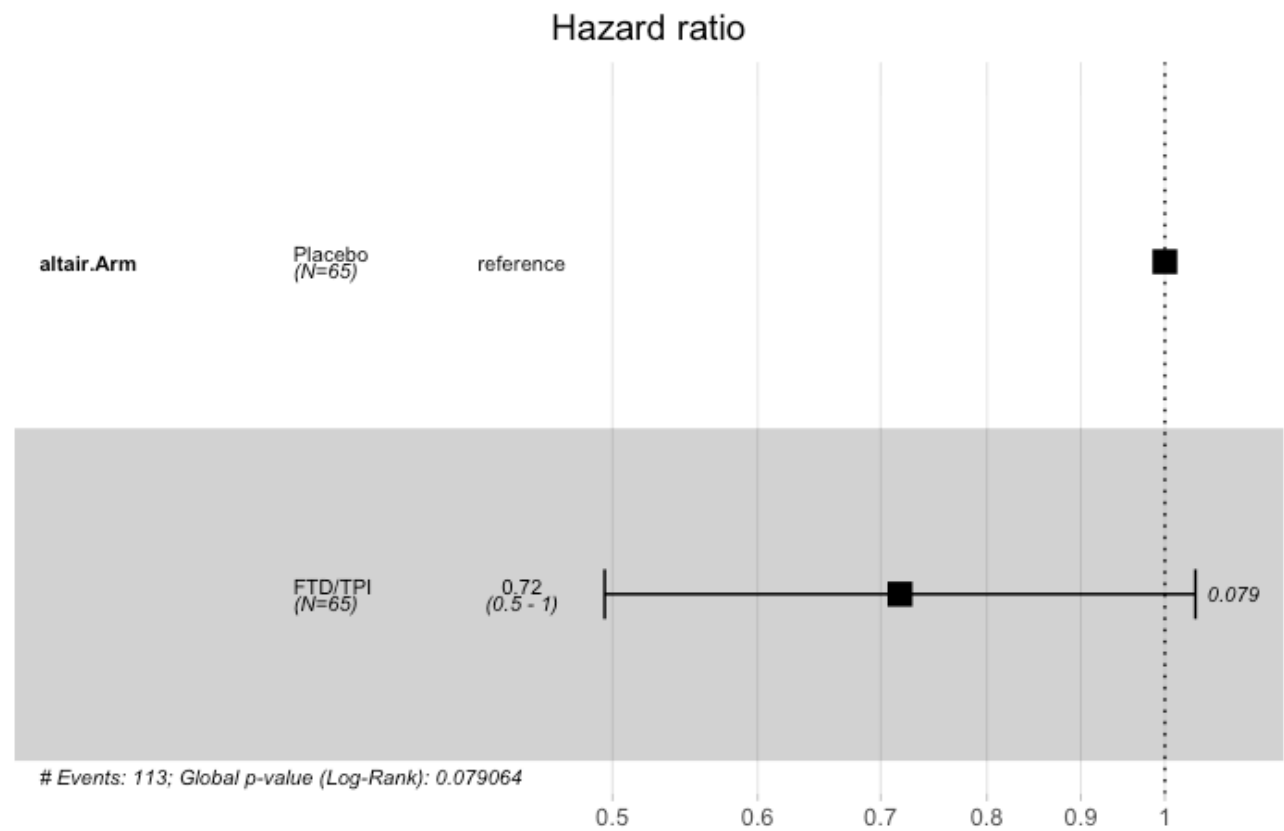
| altair.Arm=Control |        |         |          |         |        |        |       |        |
|--------------------|--------|---------|----------|---------|--------|--------|-------|--------|
| time               | n.risk | n.event | survival | std.err | lower  | 95% CI | upper | 95% CI |
| 6                  | 24     | 41      | 0.3692   | 0.0599  | 0.2540 |        | 0.485 |        |
| 12                 | 11     | 12      | 0.1813   | 0.0482  | 0.0985 |        | 0.284 |        |
| 18                 | 5      | 4       | 0.1099   | 0.0408  | 0.0464 |        | 0.204 |        |
| 24                 | 2      | 1       | 0.0824   | 0.0388  | 0.0270 |        | 0.178 |        |

| altair.Arm=Experimental |        |         |          |         |        |        |       |        |
|-------------------------|--------|---------|----------|---------|--------|--------|-------|--------|
| time                    | n.risk | n.event | survival | std.err | lower  | 95% CI | upper | 95% CI |
| 6                       | 40     | 24      | 0.630    | 0.0600  | 0.5002 |        | 0.734 |        |
| 12                      | 14     | 25      | 0.233    | 0.0534  | 0.1374 |        | 0.343 |        |
| 18                      | 8      | 4       | 0.166    | 0.0473  | 0.0863 |        | 0.269 |        |
| 24                      | 6      | 0       | 0.166    | 0.0473  | 0.0863 |        | 0.269 |        |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.Arm, data = circ\_data)

n= 130, number of events= 113

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.3323 | 0.7173    | 0.1891   | -1.757 | 0.0789 . |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.7173    | 1.394      | 0.4952    | 1.039     |

Concordance= 0.56 (se = 0.026 )  
Likelihood ratio test= 3.08 on 1 df, p=0.08  
Wald test = 3.09 on 1 df, p=0.08  
Score (logrank) test = 3.12 on 1 df, p=0.08

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.72 (0.5-1.04); p = 0.079"
```

[Hide](#)

```
#Fisher test for DFS percentages at 6, 12, 18 and 24 months
dfs_times <- c(6, 12, 18, 24)
p_values <- sapply(dfs_times, function(time) {
  neg_count <- sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  pos_count <- sum(circ_data$altair.Arm == "Placebo" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  neg_total <- sum(circ_data$altair.Arm == "FTD/TPI")
  pos_total <- sum(circ_data$altair.Arm == "Placebo")

  neg_surv <- neg_total - sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)
  pos_surv <- pos_total - sum(circ_data$altair.Arm == "Placebo" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)

  surv_matrix <- matrix(c(neg_surv, pos_surv, neg_total - neg_surv, pos_total - pos_surv), nrow = 2)
  test_result <- fisher.test(surv_matrix)
  return(test_result$p.value)
})
names(p_values) <- paste0("p-value at ", dfs_times, " months")
print(p_values)
```

| p-value at 6 months | p-value at 12 months | p-value at 18 months | p-value at 24 months |
|---------------------|----------------------|----------------------|----------------------|
| 0.004815238         | 0.522656471          | 0.466593017          | 0.320823902          |

#DFS1 by TAS vs Placebo - ctDNA negative post-surgery

[Hide](#)



```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$ctDNA1mo == "NEGATIVE",]
circ_data <- subset(circ_data, !is.na(ctDNA1mo))
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$DFS.months, event = circ\_data\$p\_evtDFS1b) ~ altair.Arm, data = circ\_data)

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 56 | 41     | 6.85   | 5.55    | 18.9    |
| altair.Arm=Experimental | 57 | 44     | 10.18  | 9.23    | 15.0    |

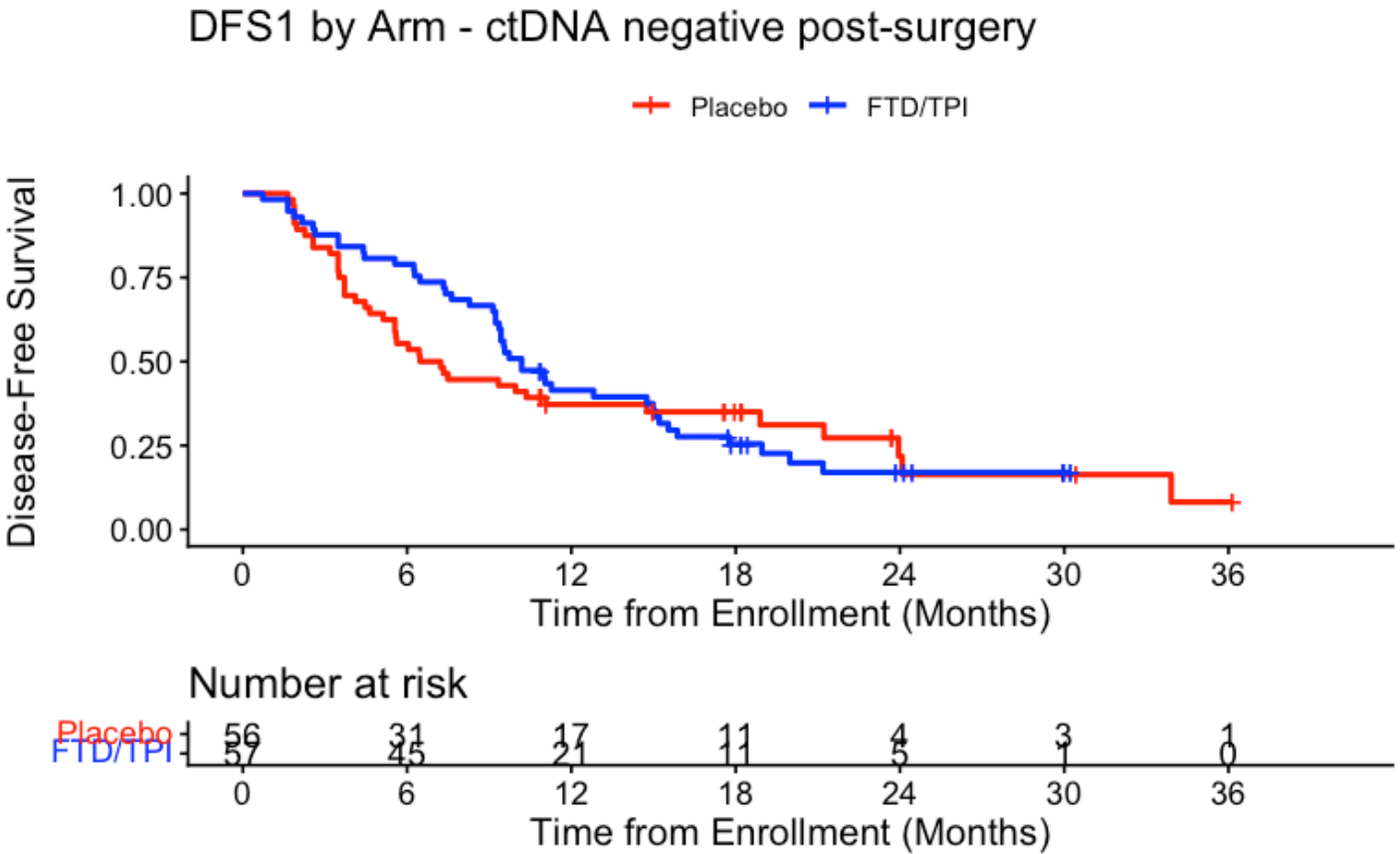
Hide

```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>  <int>   <dbl>     <dbl>
1 Control      56    41    0.732     73.2
2 Experimental  57    44    0.772     77.2
```

Hide

```
surv_object <- Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type = "log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - ctDNA negative post-surgery", ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs = c("Placebo", "FTD/TPI"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

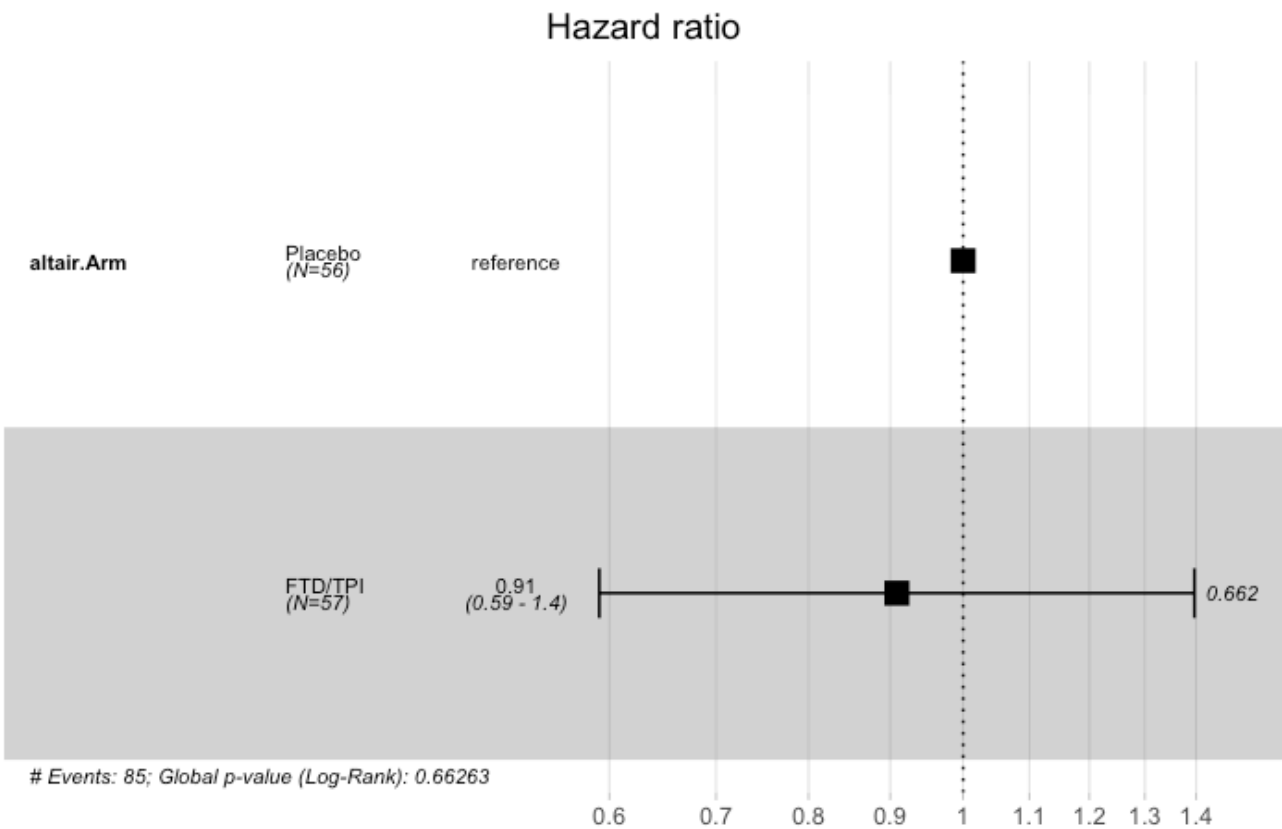
| altair.Arm=Control |        |         |          |         |              |              |  |
|--------------------|--------|---------|----------|---------|--------------|--------------|--|
| time               | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                  | 31     | 25      | 0.554    | 0.0664  | 0.4147       | 0.672        |  |
| 12                 | 17     | 10      | 0.372    | 0.0650  | 0.2472       | 0.497        |  |
| 18                 | 11     | 1       | 0.350    | 0.0648  | 0.2272       | 0.476        |  |
| 24                 | 4      | 3       | 0.218    | 0.0742  | 0.0943       | 0.374        |  |

| altair.Arm=Experimental |        |         |          |         |              |              |  |
|-------------------------|--------|---------|----------|---------|--------------|--------------|--|
| time                    | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                       | 45     | 12      | 0.789    | 0.0540  | 0.6592       | 0.875        |  |
| 12                      | 21     | 21      | 0.414    | 0.0661  | 0.2848       | 0.539        |  |
| 18                      | 11     | 8       | 0.255    | 0.0602  | 0.1471       | 0.378        |  |
| 24                      | 5      | 3       | 0.170    | 0.0567  | 0.0771       | 0.294        |  |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ altair.Arm, data = circ_data)
```

n= 113, number of events= 85

|                   | coef     | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|----------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.09568 | 0.90875   | 0.21911  | -0.437 | 0.662    |

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.9088    | 1.1        | 0.5915    | 1.396     |

Concordance= 0.538 (se = 0.03 )

Likelihood ratio test= 0.19 on 1 df, p=0.7

Wald test = 0.19 on 1 df, p=0.7

Score (logrank) test = 0.19 on 1 df, p=0.7

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 0.91 (0.59-1.4); p = 0.662"
```

#DFS2 by TAS vs Placebo - All stages

[Hide](#)

```
rm(list=ls())
```

```
setwd("~/Downloads")
```

```
circ_data <- read.csv("Altair 20240729 Dataset.csv")
```

```
circ_datadf <- as.data.frame(circ_data)
```

```
survfit(Surv(time = circ_data$DFS2.months, event = circ_data$p_evtDFS2)~altair.Arm, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$DFS2.months, event = circ\_data\$p\_evtDFS2) ~

```
altair.Arm, data = circ_data)
```

|                         | n   | events | median | 0.95LCL | 0.95UCL |
|-------------------------|-----|--------|--------|---------|---------|
| altair.Arm=Control      | 121 | 100    | 5.55   | 4.17    | 7.33    |
| altair.Arm=Experimental | 122 | 99     | 9.30   | 7.92    | 10.84   |

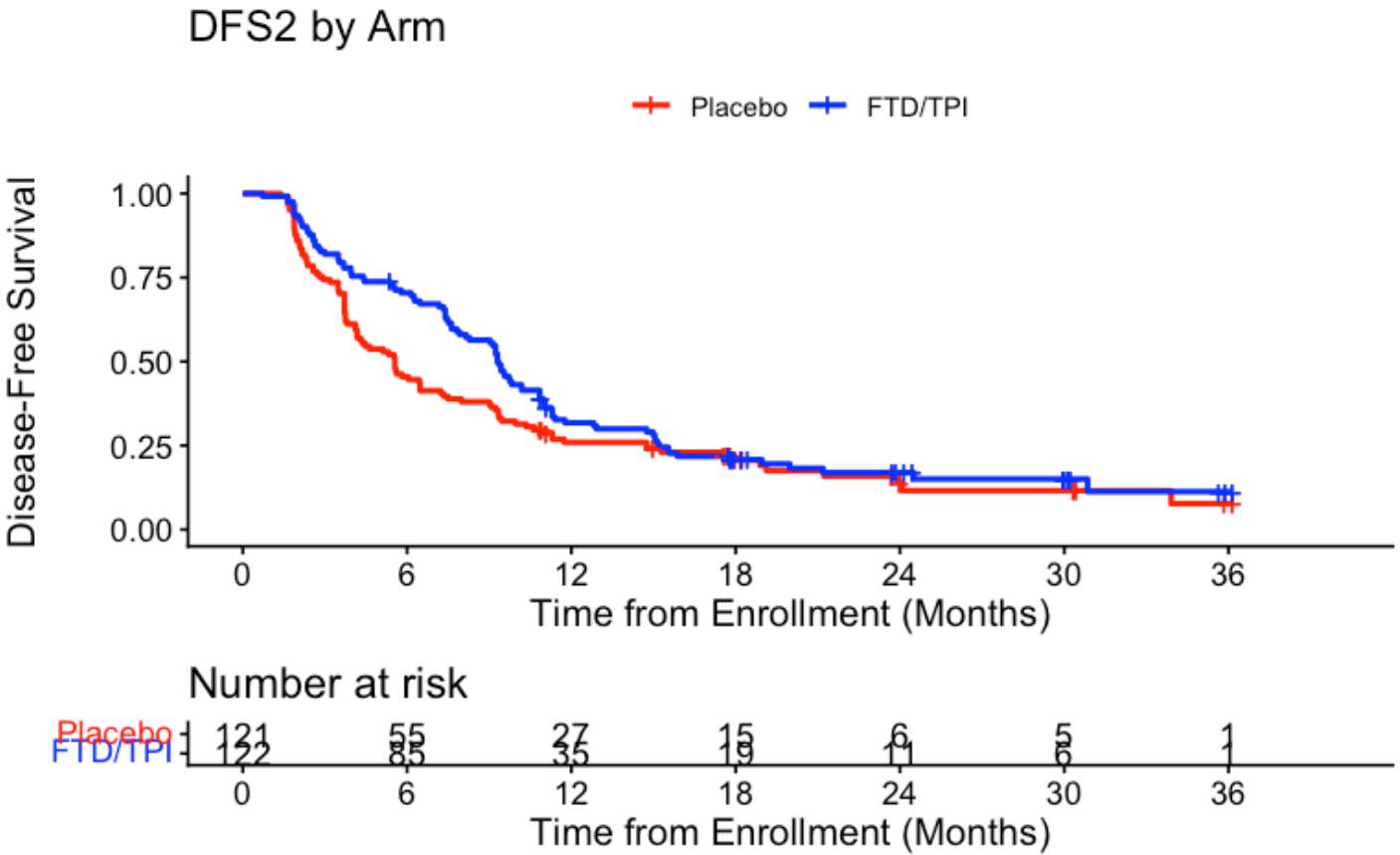
Hide

```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS2),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>  <int>    <dbl>      <dbl>
1 Control      121    100    0.826      82.6
2 Experimental 122     99    0.811      81.1
```

Hide

```
surv_object <- Surv(time = circ_data$DFS2.months, event = circ_data$p_evtDFS2)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("red","blue"), title="DFS2 by Arm", ylab= "Disease-Free Survival",
xlab="Time from Enrollment (Months)", legend.labs=c("Placebo", "FTD/TPI"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

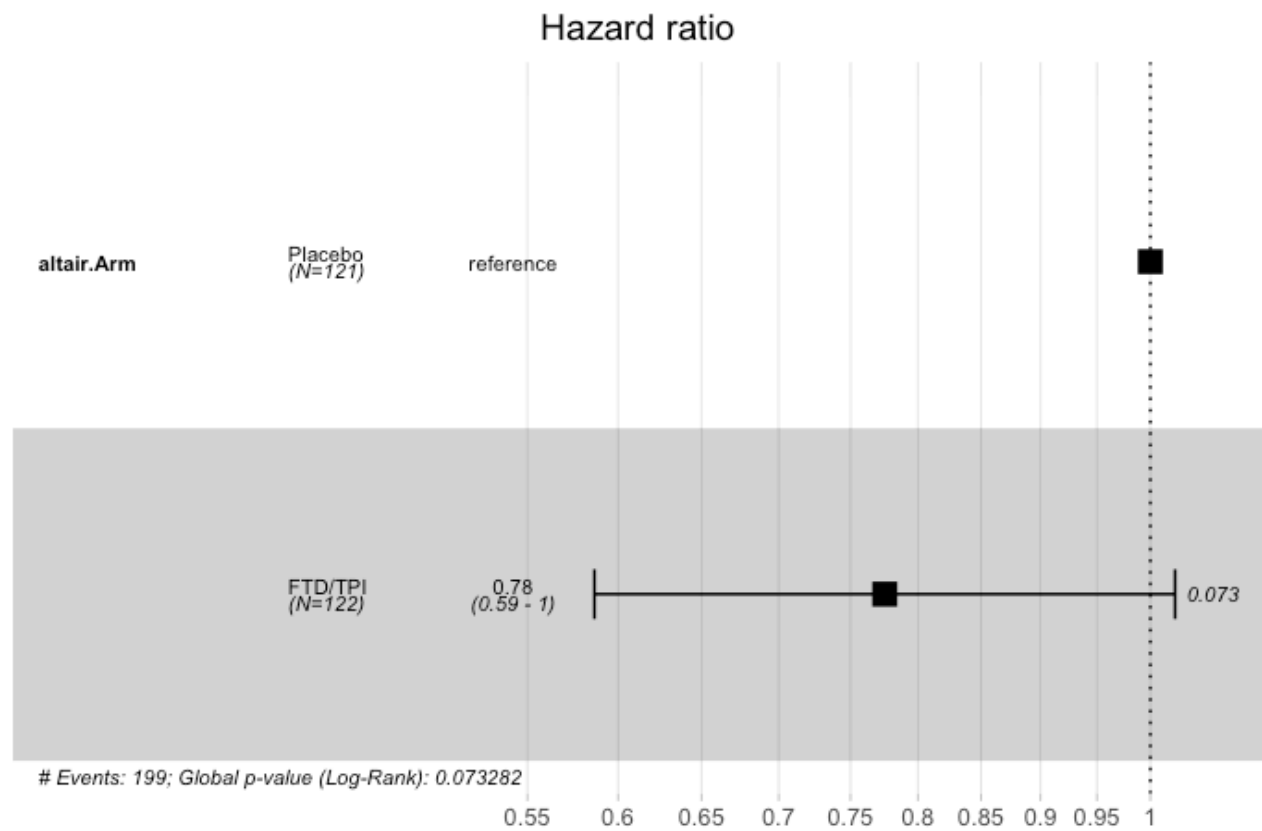
| altair.Arm=Control |        |         |          |         |        |        |       |        |
|--------------------|--------|---------|----------|---------|--------|--------|-------|--------|
| time               | n.risk | n.event | survival | std.err | lower  | 95% CI | upper | 95% CI |
| 6                  | 55     | 66      | 0.455    | 0.0453  | 0.3642 |        | 0.540 |        |
| 12                 | 27     | 23      | 0.259    | 0.0404  | 0.1842 |        | 0.341 |        |
| 18                 | 15     | 5       | 0.206    | 0.0386  | 0.1366 |        | 0.286 |        |
| 24                 | 6      | 4       | 0.139    | 0.0383  | 0.0746 |        | 0.223 |        |

| altair.Arm=Experimental |        |         |          |         |       |        |       |        |
|-------------------------|--------|---------|----------|---------|-------|--------|-------|--------|
| time                    | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
| 6                       | 85     | 36      | 0.705    | 0.0413  | 0.615 |        | 0.777 |        |
| 12                      | 35     | 46      | 0.318    | 0.0428  | 0.236 |        | 0.402 |        |
| 18                      | 19     | 12      | 0.208    | 0.0380  | 0.139 |        | 0.287 |        |
| 24                      | 11     | 3       | 0.169    | 0.0370  | 0.104 |        | 0.248 |        |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.Arm, data = circ\_data)

n= 243, number of events= 199

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.2549 | 0.7750    | 0.1421   | -1.793 | 0.073 .  |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.775     | 1.29       | 0.5866    | 1.024     |

Concordance= 0.552 (se = 0.019 )  
Likelihood ratio test= 3.21 on 1 df, p=0.07  
Wald test = 3.22 on 1 df, p=0.07  
Score (logrank) test = 3.23 on 1 df, p=0.07

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.78 (0.59-1.02); p = 0.073"
```

#DFS1 by ctDNA MRD enrollment timepoint TAS vs Placebo

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$p_AltBaselineWin=="MRD",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
~
      altair.Arm, data = circ_data)
```

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 30 | 26     | 4.17   | 3.71    | 7.98    |
| altair.Arm=Experimental | 28 | 22     | 7.77   | 5.39    | 11.30   |

Hide

```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

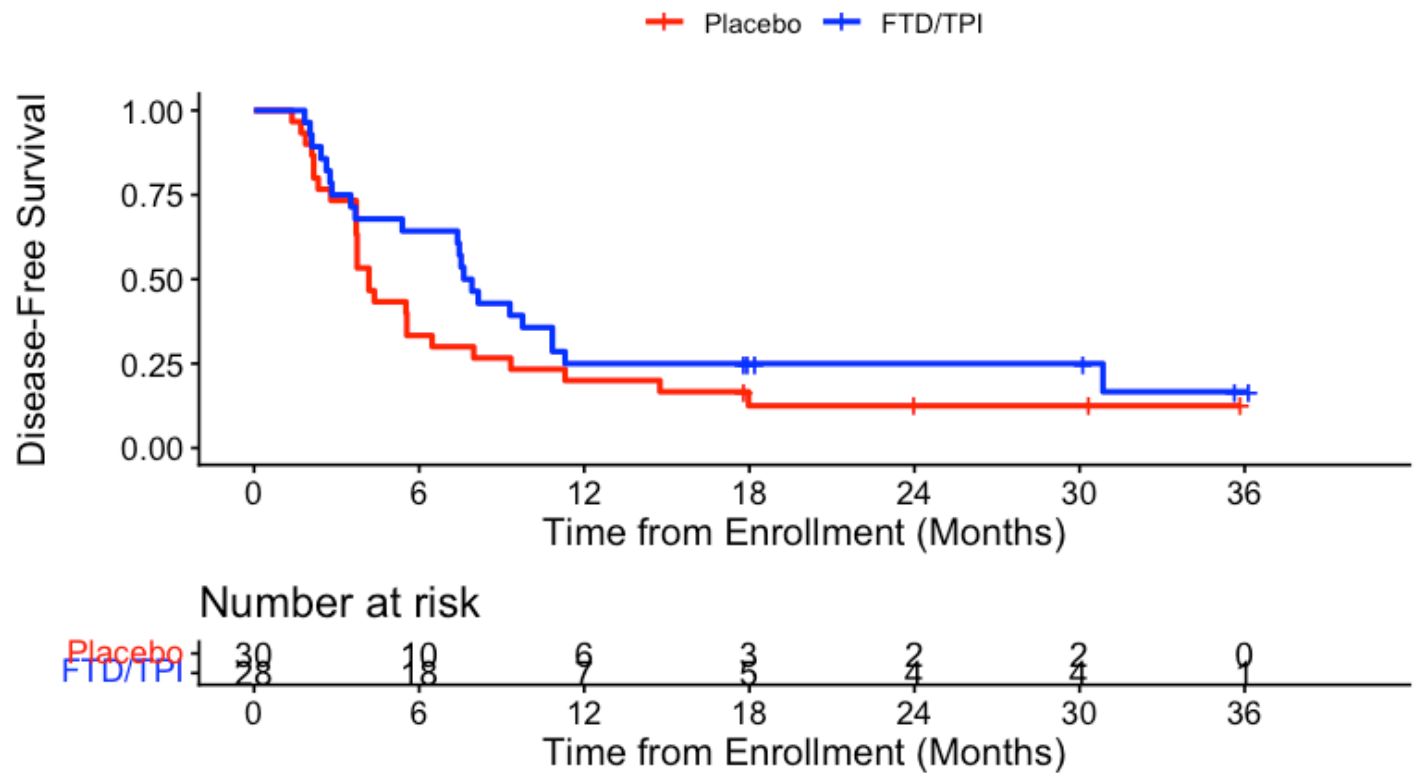


```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>   <int>   <dbl>     <dbl>
1 Control      30     26    0.867     86.7
2 Experimental 28     22    0.786     78.6
```

[Hide](#)

```
surv_object <-Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - ctDNA MRD Enrollment",
ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("Placebo", "FTD/TPI"), legend.title="")
```

DFS1 by Arm - ctDNA MRD Enrollment



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

Call: `survfit(formula = surv_object ~ altair.Arm, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

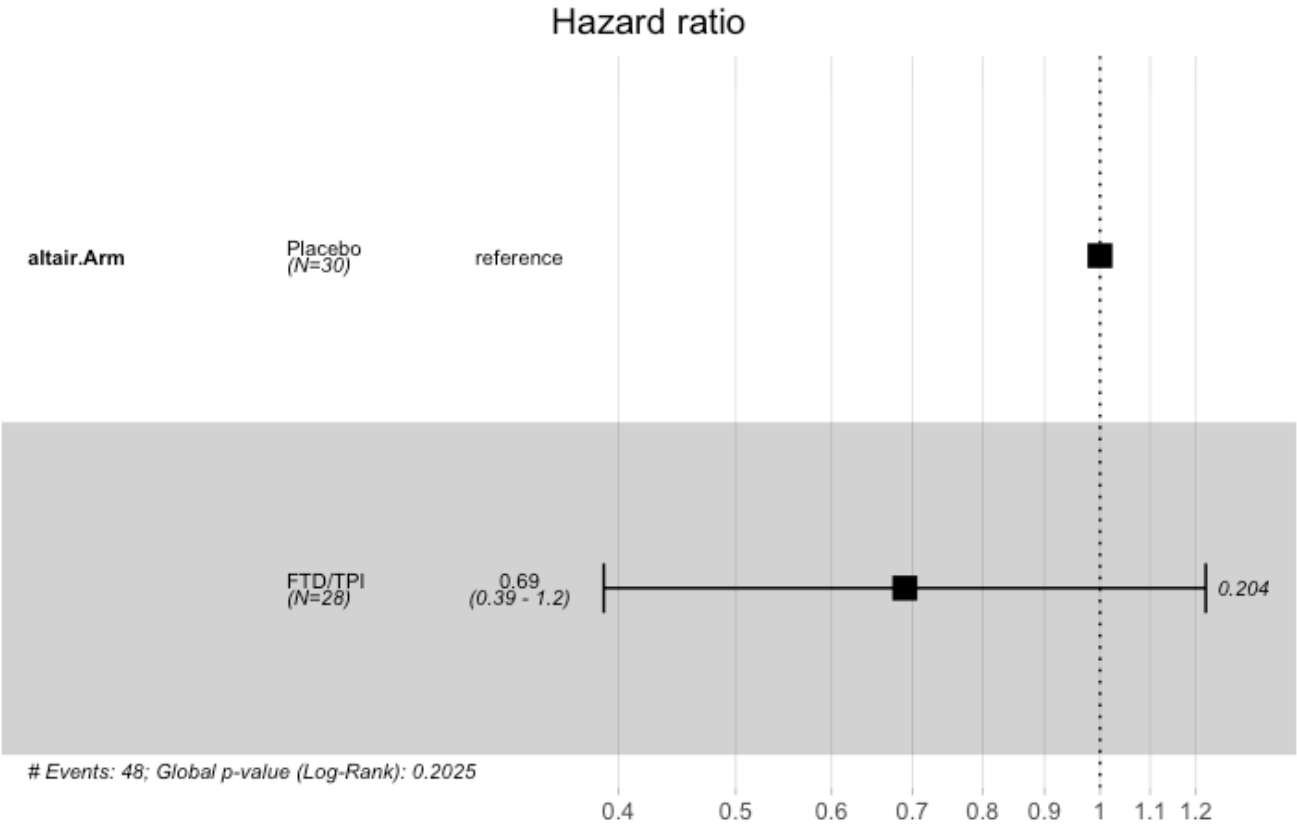
| altair.Arm=Control |        |         |          |         |        |        |       |        |
|--------------------|--------|---------|----------|---------|--------|--------|-------|--------|
| time               | n.risk | n.event | survival | std.err | lower  | 95% CI | upper | 95% CI |
| 6                  | 10     | 20      | 0.333    | 0.0861  | 0.1753 |        | 0.500 |        |
| 12                 | 6      | 4       | 0.200    | 0.0730  | 0.0812 |        | 0.356 |        |
| 18                 | 3      | 2       | 0.125    | 0.0625  | 0.0357 |        | 0.273 |        |
| 24                 | 2      | 0       | 0.125    | 0.0625  | 0.0357 |        | 0.273 |        |

| altair.Arm=Experimental |        |         |          |         |       |        |       |        |
|-------------------------|--------|---------|----------|---------|-------|--------|-------|--------|
| time                    | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
| 6                       | 18     | 10      | 0.643    | 0.0906  | 0.438 |        | 0.789 |        |
| 12                      | 7      | 11      | 0.250    | 0.0818  | 0.111 |        | 0.418 |        |
| 18                      | 5      | 0       | 0.250    | 0.0818  | 0.111 |        | 0.418 |        |
| 24                      | 4      | 0       | 0.250    | 0.0818  | 0.111 |        | 0.418 |        |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.Arm, data = circ\_data)

n= 58, number of events= 48

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.3714 | 0.6897    | 0.2921   | -1.271 | 0.204    |

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.6897    | 1.45       | 0.3891    | 1.223     |

Concordance= 0.555 (se = 0.04 )  
Likelihood ratio test= 1.62 on 1 df, p=0.2  
Wald test = 1.62 on 1 df, p=0.2  
Score (logrank) test = 1.63 on 1 df, p=0.2

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 0.69 (0.39-1.22); p = 0.204"
```

[Hide](#)

```
#Fisher test for DFS percentages at 6, 12, 18 and 24 months
dfs_times <- c(6, 12, 18, 24)
p_values <- sapply(dfs_times, function(time) {
  neg_count <- sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  pos_count <- sum(circ_data$altair.Arm == "Placebo" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  neg_total <- sum(circ_data$altair.Arm == "FTD/TPI")
  pos_total <- sum(circ_data$altair.Arm == "Placebo")

  neg_surv <- neg_total - sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)
  pos_surv <- pos_total - sum(circ_data$altair.Arm == "Placebo" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)

  surv_matrix <- matrix(c(neg_surv, pos_surv, neg_total - neg_surv, pos_total - pos_surv), nrow = 2)
  test_result <- fisher.test(surv_matrix)
  return(test_result$p.value)
})
names(p_values) <- paste0("p-value at ", dfs_times, " months")
print(p_values)
```

| p-value at 6 months | p-value at 12 months | p-value at 18 months | p-value at 24 months |
|---------------------|----------------------|----------------------|----------------------|
| 0.03454285          | 0.75694899           | 0.32484791           | 0.32484791           |

#DFS1 by ctDNA On-treatment timepoint TAS vs Placebo

[Hide](#)

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$p_AltBaselineWin=="OnTreatment",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$DFS.months, event = circ\_data\$p\_evtDFS1b) ~ altair.Arm, data = circ\_data)

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 19 | 16     | 9.07   | 4.30    | 17.5    |
| altair.Arm=Experimental | 14 | 13     | 6.64   | 2.99    | NA      |

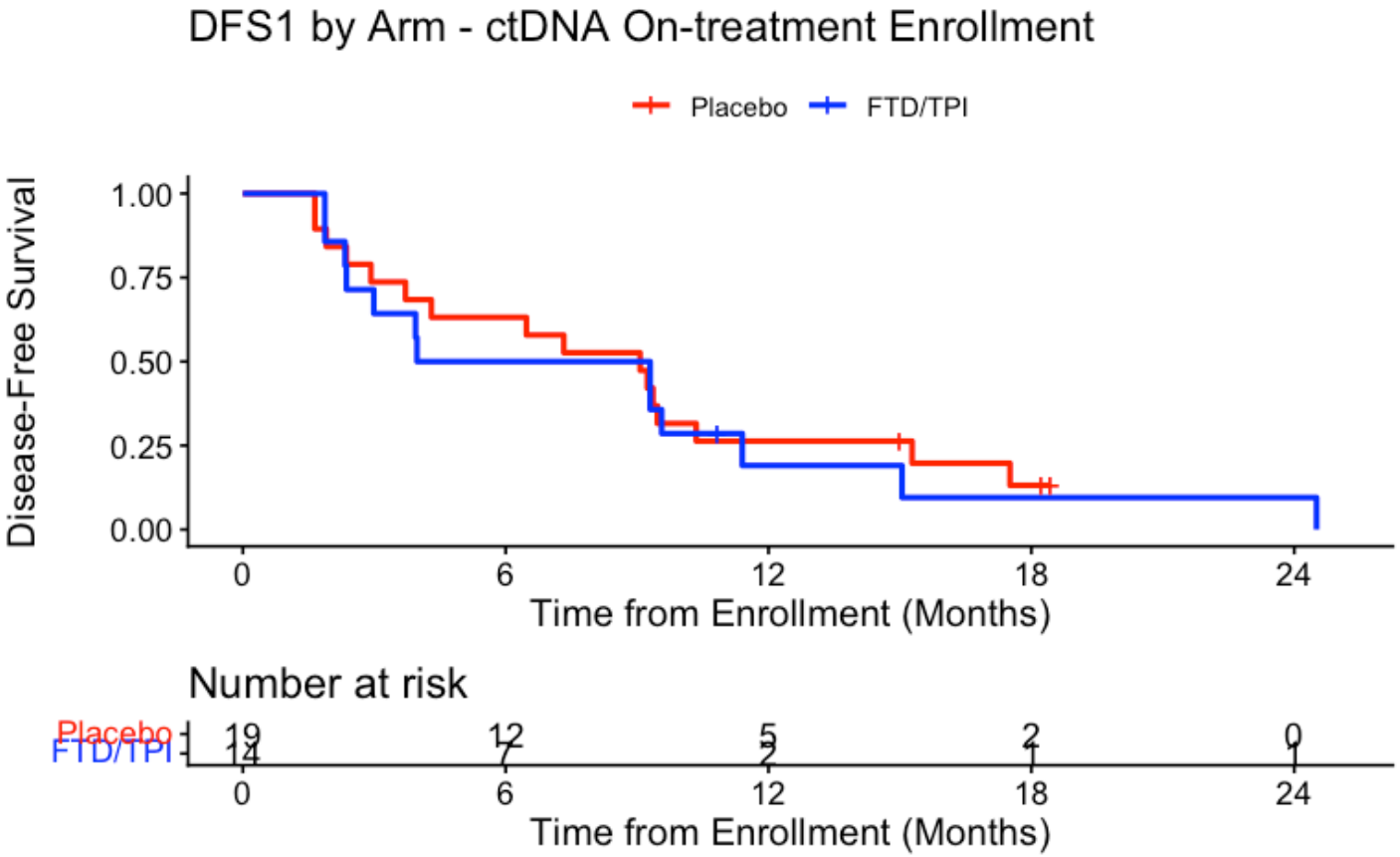
Hide

```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>  <int>   <dbl>      <dbl>
1 Control         19     16  0.842      84.2
2 Experimental    14     13  0.929      92.9
```

Hide

```
surv_object <- Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type = "log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - ctDNA On-treatment Enrollment", ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("Placebo", "FTD/TPI"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

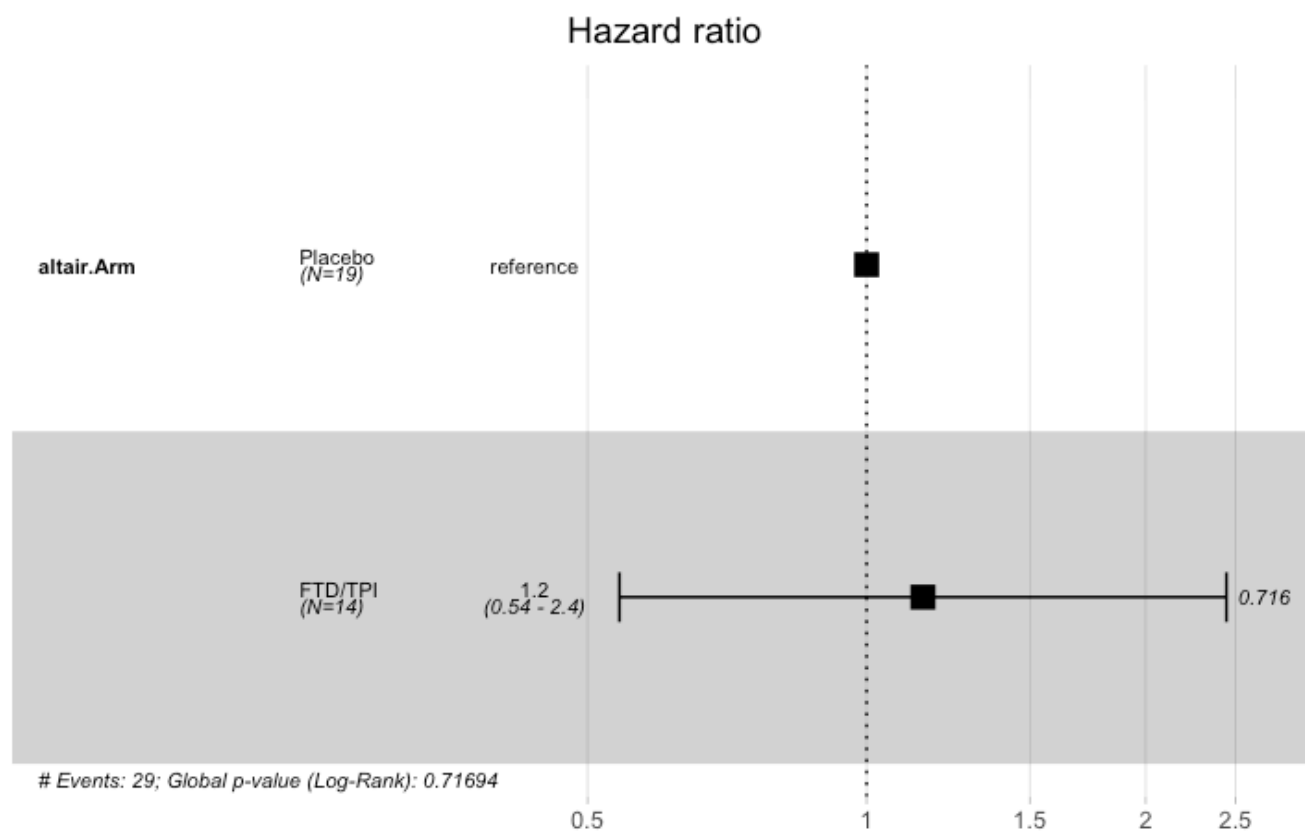
```
Call: survfit(formula = surv_object ~ altair.Arm, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

```
altair.Arm=Control
time n.risk n.event survival std.err lower 95% CI upper 95% CI
  6    12     7    0.632  0.1107    0.3790    0.804
 12     5     7    0.263  0.1010    0.0958    0.468
 18     2     2    0.132  0.0829    0.0240    0.332
```

```
altair.Arm=Experimental
time n.risk n.event survival std.err lower 95% CI upper 95% CI
  6     7     7    0.5000  0.1336    0.22859    0.722
 12     2     4    0.1905  0.1119    0.03612    0.437
 18     1     1    0.0952  0.0876    0.00635    0.335
 24     1     0    0.0952  0.0876    0.00635    0.335
```

[Hide](#)

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
  labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```


[Hide](#)

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ altair.Arm, data = circ_data)
```

n= 33, number of events= 29

|                   | coef   | exp(coef) | se(coef) | z     | Pr(> z ) |
|-------------------|--------|-----------|----------|-------|----------|
| altair.ArmFTD/TPI | 0.1399 | 1.1502    | 0.3846   | 0.364 | 0.716    |

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 1.15      | 0.8694     | 0.5413    | 2.444     |

Concordance= 0.516 (se = 0.053 )

Likelihood ratio test= 0.13 on 1 df, p=0.7

Wald test = 0.13 on 1 df, p=0.7

Score (logrank) test = 0.13 on 1 df, p=0.7

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C  
I, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 1.15 (0.54-2.44); p = 0.716"
```

[Hide](#)



```
#Fisher test for DFS percentages at 6, 12, 18 and 24 months
dfs_times <- c(6, 12, 18, 24)
p_values <- sapply(dfs_times, function(time) {
  neg_count <- sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  pos_count <- sum(circ_data$altair.Arm == "Placebo" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  neg_total <- sum(circ_data$altair.Arm == "FTD/TPI")
  pos_total <- sum(circ_data$altair.Arm == "Placebo")

  neg_surv <- neg_total - sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)
  pos_surv <- pos_total - sum(circ_data$altair.Arm == "Placebo" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)

  surv_matrix <- matrix(c(neg_surv, pos_surv, neg_total - neg_surv, pos_total - pos_surv), nrow = 2)
  test_result <- fisher.test(surv_matrix)
  return(test_result$p.value)
})
names(p_values) <- paste0("p-value at ", dfs_times, " months")
print(p_values)
```

| p-value at 6 months | p-value at 12 months | p-value at 18 months | p-value at 24 months |
|---------------------|----------------------|----------------------|----------------------|
| 0.4969359           | 1.0000000            | 1.0000000            | 1.0000000            |

#DFS1 by ctDNA Surveillance timepoint TAS vs Placebo

[Hide](#)

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$p_AltBaselineWin=="Surveillance",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.Arm, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b) ~
  altair.Arm, data = circ_data)
```

|                         | n  | events | median | 0.95LCL | 0.95UCL |
|-------------------------|----|--------|--------|---------|---------|
| altair.Arm=Control      | 72 | 57     | 5.57   | 4.11    | 9.33    |
| altair.Arm=Experimental | 80 | 64     | 9.73   | 9.13    | 11.76   |

[Hide](#)

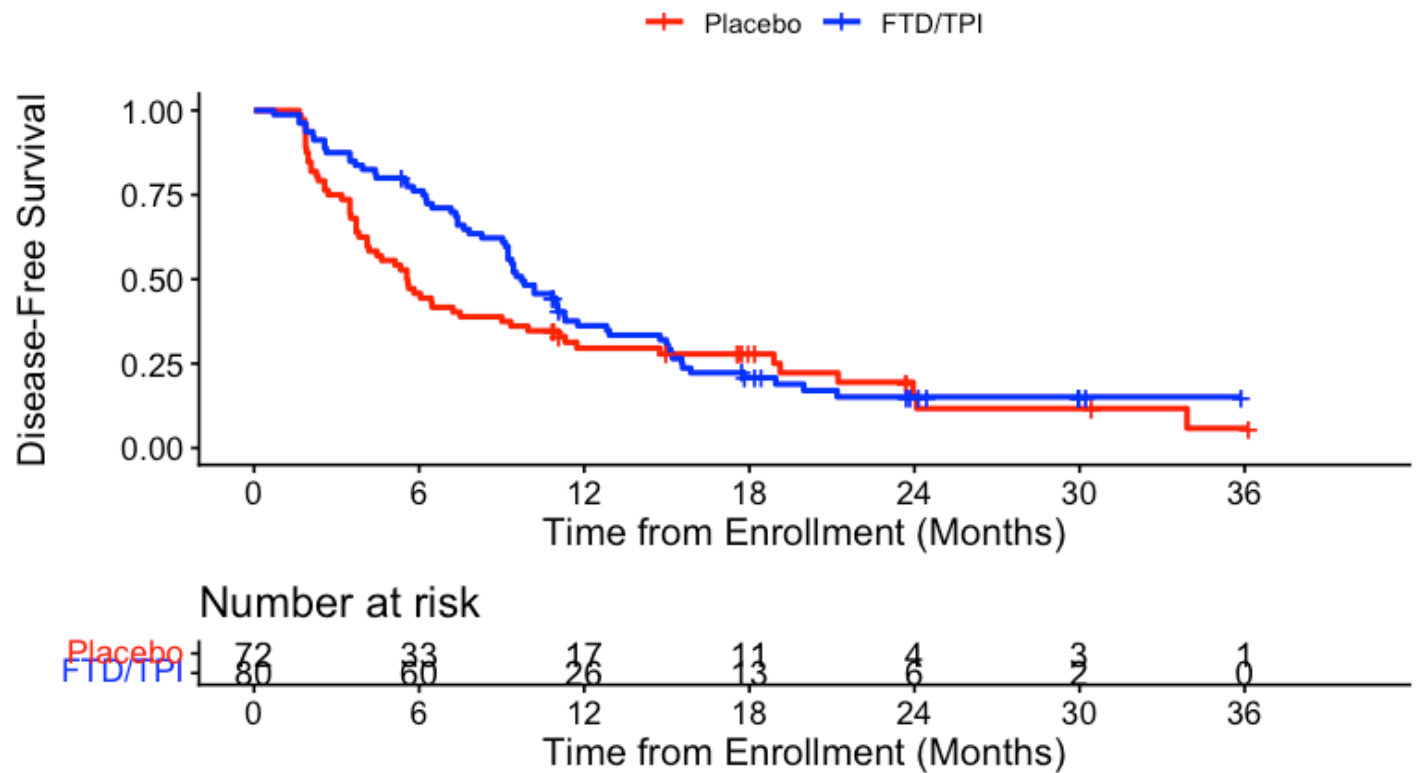
```
event_summary <- circ_data %>%
  group_by(altair.Arm) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 2 × 5
  altair.Arm   Total Events Fraction Percentage
  <chr>       <int>  <int>    <dbl>     <dbl>
1 Control         72    57    0.792     79.2
2 Experimental    80    64    0.8       80
```

[Hide](#)

```
surv_object <- Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int=0.95, conf.type
="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("red","blue"), title="DFS1 by Arm - ctDNA Surveillance Enrollment",
ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labels=c("Placebo", "FTD/TPI"), legend.title="")
```

DFS1 by Arm - ctDNA Surveillance Enrollment



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

Call: `survfit(formula = surv_object ~ altair.Arm, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

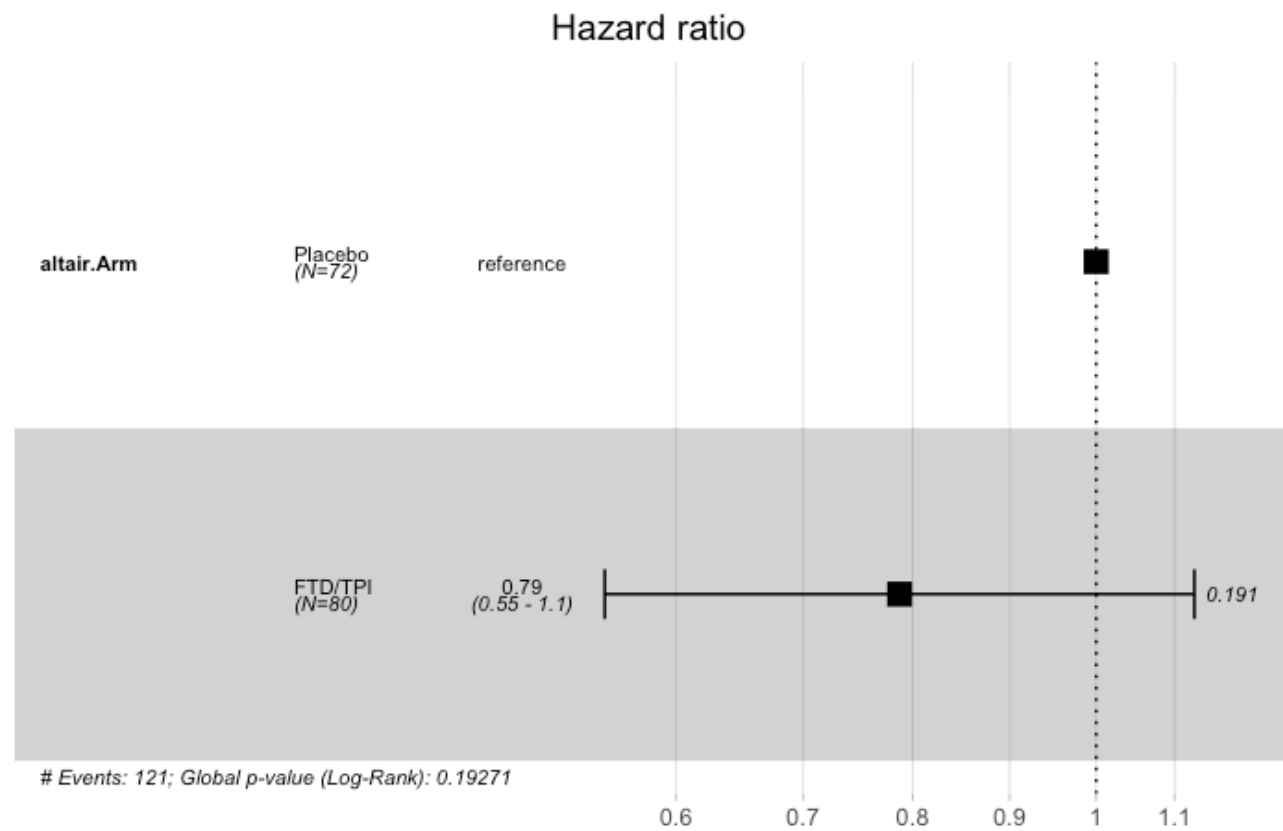
| altair.Arm=Control |        |         |          |         |        |              |        |
|--------------------|--------|---------|----------|---------|--------|--------------|--------|
| time               | n.risk | n.event | survival | std.err | lower  | 95% CI upper | 95% CI |
| 6                  | 33     | 39      | 0.458    | 0.0587  | 0.3408 |              | 0.568  |
| 12                 | 17     | 11      | 0.296    | 0.0551  | 0.1933 |              | 0.406  |
| 18                 | 11     | 1       | 0.278    | 0.0545  | 0.1780 |              | 0.388  |
| 24                 | 4      | 4       | 0.156    | 0.0565  | 0.0657 |              | 0.281  |

| altair.Arm=Experimental |        |         |          |         |        |              |        |
|-------------------------|--------|---------|----------|---------|--------|--------------|--------|
| time                    | n.risk | n.event | survival | std.err | lower  | 95% CI upper | 95% CI |
| 6                       | 60     | 19      | 0.762    | 0.0477  | 0.6525 |              | 0.841  |
| 12                      | 26     | 31      | 0.362    | 0.0547  | 0.2568 |              | 0.468  |
| 18                      | 13     | 11      | 0.208    | 0.0473  | 0.1243 |              | 0.307  |
| 24                      | 6      | 3       | 0.151    | 0.0443  | 0.0774 |              | 0.248  |

Hide

```
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
labels = c("Placebo", "FTD/TPI"))
cox_fit <- coxph(surv_object ~ altair.Arm, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.Arm, data = circ\_data)

n= 152, number of events= 121

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.2391 | 0.7873    | 0.1829   | -1.307 | 0.191    |

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.7873    | 1.27       | 0.5501    | 1.127     |

Concordance= 0.559 (se = 0.025 )  
Likelihood ratio test= 1.7 on 1 df, p=0.2  
Wald test = 1.71 on 1 df, p=0.2  
Score (logrank) test = 1.72 on 1 df, p=0.2

Hide

```

cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)

```

```
[1] "HR = 0.79 (0.55-1.13); p = 0.191"
```

Hide

```

#Fisher test for DFS percentages at 6, 12, 18 and 24 months
dfs_times <- c(6, 12, 18, 24)
p_values <- sapply(dfs_times, function(time) {
  neg_count <- sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  pos_count <- sum(circ_data$altair.Arm == "Placebo" & circ_data$DFS.months >= time & circ_data$p_evtDFS1b == 0)
  neg_total <- sum(circ_data$altair.Arm == "FTD/TPI")
  pos_total <- sum(circ_data$altair.Arm == "Placebo")

  neg_surv <- neg_total - sum(circ_data$altair.Arm == "FTD/TPI" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)
  pos_surv <- pos_total - sum(circ_data$altair.Arm == "Placebo" & circ_data$p_evtDFS1b == 1 & circ_data$DFS.months < time)

  surv_matrix <- matrix(c(neg_surv, pos_surv, neg_total - neg_surv, pos_total - pos_surv), nrow = 2)
  test_result <- fisher.test(surv_matrix)
  return(test_result$p.value)
})
names(p_values) <- paste0("p-value at ", dfs_times, " months")
print(p_values)

```

| p-value at 6 months | p-value at 12 months | p-value at 18 months | p-value at 24 months |
|---------------------|----------------------|----------------------|----------------------|
| 0.0001278464        | 0.3959706440         | 0.4666619773         | 0.6942842858         |

#Barplot with enrollment timepoint at any time by Arm

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")

circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels=c("Control","Experimental"),
labels = c("Placebo", "FTD/TPI"))
circ_data$p_AltBaselineWin <- factor(circ_data$p_AltBaselineWin, levels = c("MRD", "OnTreatment", "Surveillance"), labels = c("MRD", "On Treatment", "Surveillance"))
contingency_table <- table(circ_data$altair.Arm, circ_data$p_AltBaselineWin)
chi_square_test <- chisq.test(contingency_table)
print(chi_square_test)
```

### Pearson's Chi-squared test

```
data: contingency_table
X-squared = 1.2435, df = 2, p-value = 0.537
```

[Hide](#)

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

### Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.5471
alternative hypothesis: two.sided
```

[Hide](#)

```
print(contingency_table)
```

|         | MRD | On Treatment | Surveillance |
|---------|-----|--------------|--------------|
| Placebo | 30  | 19           | 72           |
| FTD/TPI | 28  | 14           | 80           |

[Hide](#)

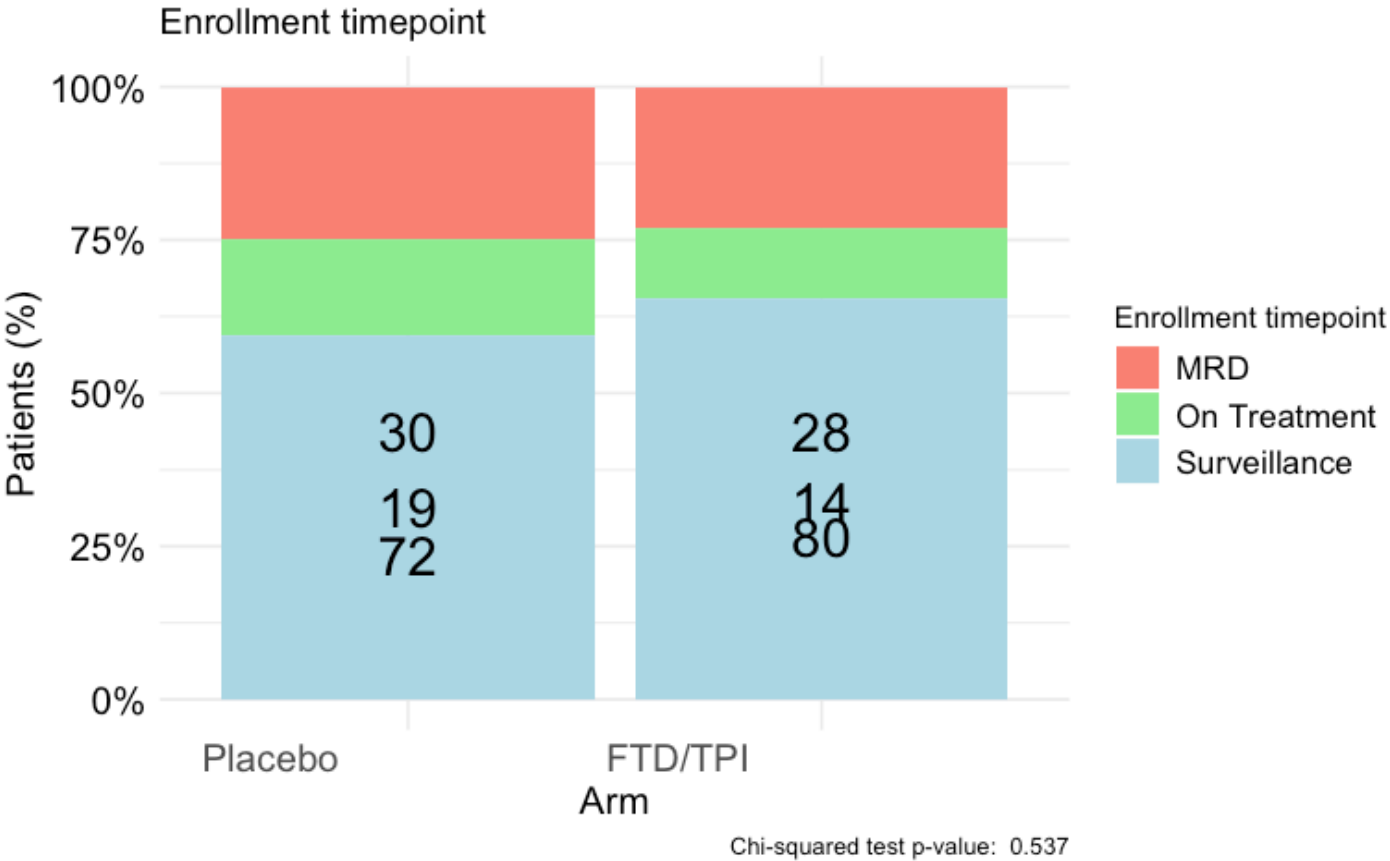
```
p_values <- c(chi_square_test$p.value, fisher_exact_test$p.value)
p_adjusted <- p.adjust(p_values, method = "bonferroni")
names(p_adjusted) <- c("Chi-Square Test", "Fisher's Exact Test")
print(p_adjusted)
```

| Chi-Square Test | Fisher's Exact Test |
|-----------------|---------------------|
| 1               | 1                   |

```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "Enrollment timepoint",
        x = "Arm",
        y = "Patients (%)",
        fill = "Enrollment timepoint",
        caption = paste("Chi-squared test p-value: ", format.pval(chi_square_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Surveillance" = "lightblue", "On Treatment" = "lightgreen", "MRD" = "salmon")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
        legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



Hide

```
#Calculate median MTM/mL for enrollment timepoint
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
result <- circ_data %>%
  group_by(p_AltBaselineWin) %>%
  summarise(
    Median = median(p_AltBaselineMTM, na.rm = TRUE),
    Range = paste(min(p_AltBaselineMTM, na.rm = TRUE), max(p_AltBaselineMTM, na.rm = TRUE), sep = " - ")
  )
print(result)
```

```
# A tibble: 3 × 3
  p_AltBaselineWin Median Range
  <chr>           <dbl> <chr>
1 MRD             0.972 0.026508782 - 121.2686353
2 OnTreatment     0.822 0.06015441805 - 166.3342558
3 Surveillance    0.240 0.02627334075 - 250.47
```

#Enrollment MTM/mL by ctDNA clearance in TAS-102 vs Placebo Arms

Hide



```
#Placebo
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$altair.group.SAP_MSv2!="1b = Exclude: No on-Tx TPs",]
circ_data <- circ_data[circ_data$altair.Arm=="Control",]

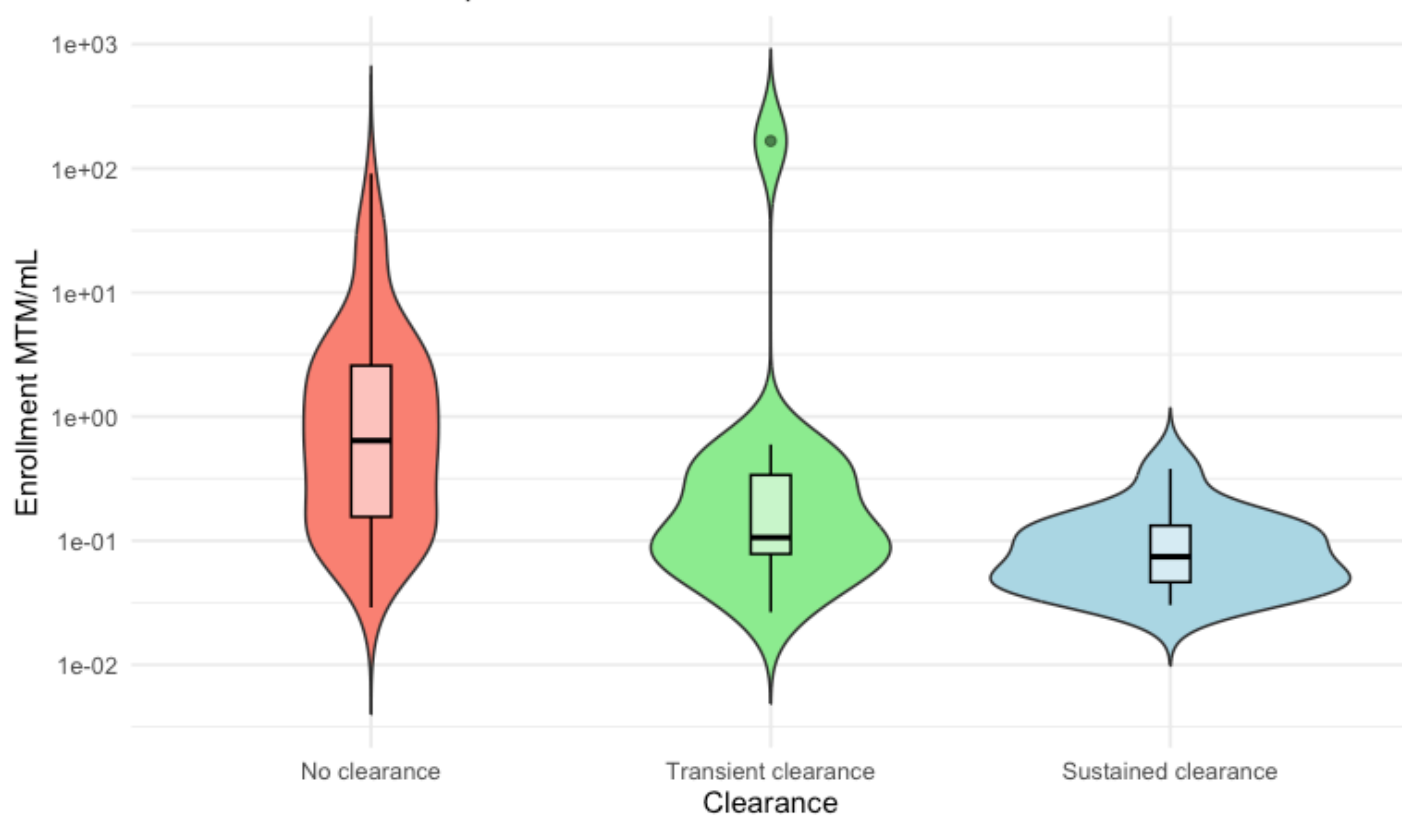
# Transform p_MRD_MTM with log10
circ_data$p_AltBaselineMTM <- as.numeric(as.character(circ_data$p_AltBaselineMTM))
circ_data$p_evtDFS1b <- factor(circ_data$p_evtDFS1b, levels=c("TRUE","FALSE"), labels =
c("Recurrence", "No Recurrence"))
circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("No clearanc
e", "Transient clearance", "Sustained clearance"))
median_p_MRD_MTM <- aggregate(p_AltBaselineMTM ~ altair.resultPatW, data = circ_data, FU
N = median)
print(median_p_MRD_MTM)
```

|   | altair.resultPatW   | p_AltBaselineMTM |
|---|---------------------|------------------|
| 1 | No clearance        | 0.63992898       |
| 2 | Transient clearance | 0.10612747       |
| 3 | Sustained clearance | 0.07446615       |

[Hide](#)

```
# Create violin plot with log10 scale on y-axis
ggplot(circ_data, aes(x=altair.resultPatW, y=p_AltBaselineMTM, fill=altair.resultPatW))
+
  geom_violin(trim=FALSE) +
  scale_fill_manual(values=c("Sustained clearance"="lightblue", "Transient clearance"="l
ightgreen", "No clearance"="salmon")) +
  geom_boxplot(width=0.1, fill="white", colour="black", alpha=0.5) +
  scale_y_log10(breaks=c(0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000)) +
  labs(title="Enrollment MTM/mL | Clearance - Placebo Arm", x="Clearance", y="Enrollment
MTM/mL") +
  theme_minimal() +
  theme(legend.position="none")
```

## Enrollment MTM/mL | Clearance - Placebo Arm



Hide

```
m3_1v2 <- wilcox.test(p_AltBaselineMTM ~ altair.resultPatW,
                      data = circ_data[circ_data$altair.resultPatW %in% c("Sustained cle
arance", "Transient clearance"), ],
                      na.rm = TRUE)
print(m3_1v2)
```

Wilcoxon rank sum exact test

data: p\_AltBaselineMTM by altair.resultPatW  
W = 153, p-value = 0.09254  
alternative hypothesis: true location shift is not equal to 0

Hide

```
m3_1v3 <- wilcox.test(p_AltBaselineMTM ~ altair.resultPatW,
                      data = circ_data[circ_data$altair.resultPatW %in% c("Sustained cle
arance", "No clearance"), ],
                      na.rm = TRUE)
print(m3_1v3)
```

Wilcoxon rank sum test with continuity correction

```
data:  p_AltBaselineMTM by altair.resultPatW
W = 1053, p-value = 4.344e-06
alternative hypothesis: true location shift is not equal to 0
```

[Hide](#)

```
m3_2v3 <- wilcox.test(p_AltBaselineMTM ~ altair.resultPatW,
                      data = circ_data[circ_data$altair.resultPatW %in% c("Transient clearance", "No clearance"), ],
                      na.rm = TRUE)
print(m3_2v3)
```

Wilcoxon rank sum test with continuity correction

```
data:  p_AltBaselineMTM by altair.resultPatW
W = 1010, p-value = 0.002179
alternative hypothesis: true location shift is not equal to 0
```

[Hide](#)

```
# Create a table with p-values
p_value_table <- data.frame(
  Comparison = c("Sustained vs Transient", "Sustained vs No Clearance", "Transient vs No Clearance"),
  P_Value = c(m3_1v2$p.value, m3_1v3$p.value, m3_2v3$p.value)
)
print(p_value_table)
```

|   | Comparison                | P_Value      |
|---|---------------------------|--------------|
| 1 | Sustained vs Transient    | 9.253992e-02 |
| 2 | Sustained vs No Clearance | 4.344491e-06 |
| 3 | Transient vs No Clearance | 2.179474e-03 |

[Hide](#)

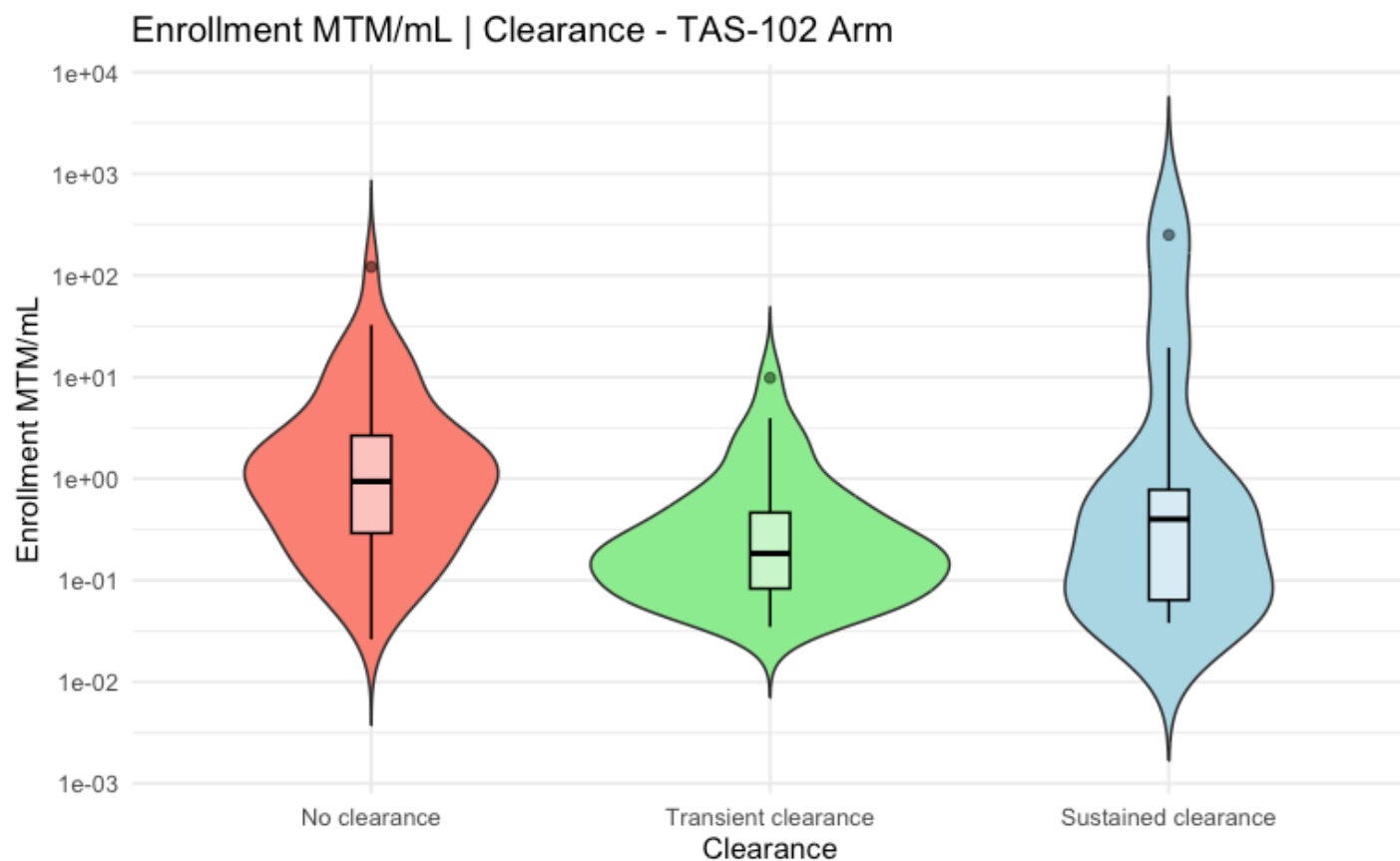
```
#TAS-102
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$altair.group.SAP_MSv2!="1b = Exclude: No on-Tx TPs",]
circ_data <- circ_data[circ_data$altair.Arm=="Experimental",]

# Transform p_MRD_MTM with log10
circ_data$p_AltBaselineMTM <- as.numeric(as.character(circ_data$p_AltBaselineMTM))
circ_data$p_evtDFS1b <- factor(circ_data$p_evtDFS1b, levels=c("TRUE","FALSE"), labels =
c("Recurrence", "No Recurrence"))
circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("No clearanc
e", "Transient clearance", "Sustained clearance"))
median_p_MRD_MTM <- aggregate(p_AltBaselineMTM ~ altair.resultPatW, data = circ_data, FU
N = median)
print(median_p_MRD_MTM)
```

|   | altair.resultPatW   | p_AltBaselineMTM |
|---|---------------------|------------------|
| 1 | No clearance        | 0.9419575        |
| 2 | Transient clearance | 0.1840031        |
| 3 | Sustained clearance | 0.4000000        |

[Hide](#)

```
# Create violin plot with log10 scale on y-axis
ggplot(circ_data, aes(x=altair.resultPatW, y=p_AltBaselineMTM, fill=altair.resultPatW))
+
  geom_violin(trim=FALSE) +
  scale_fill_manual(values=c("Sustained clearance"="lightblue", "Transient clearance"="l
ightgreen", "No clearance"="salmon")) +
  geom_boxplot(width=0.1, fill="white", colour="black", alpha=0.5) +
  scale_y_log10(breaks=c(0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000)) +
  labs(title="Enrollment MTM/mL | Clearance - TAS-102 Arm", x="Clearance", y="Enrollment
MTM/mL") +
  theme_minimal() +
  theme(legend.position="none")
```


[Hide](#)

```
m3_1v2 <- wilcox.test(p_AltBaselineMTM ~ altair.resultPatW,
                      data = circ_data[circ_data$altair.resultPatW %in% c("Sustained cle
arance", "Transient clearance"), ],
                      na.rm = TRUE)
print(m3_1v2)
```

Wilcoxon rank sum exact test

data: p\_AltBaselineMTM by altair.resultPatW  
W = 241, p-value = 0.7395  
alternative hypothesis: true location shift is not equal to 0

[Hide](#)

```
m3_1v3 <- wilcox.test(p_AltBaselineMTM ~ altair.resultPatW,
                      data = circ_data[circ_data$altair.resultPatW %in% c("Sustained cle
arance", "No clearance"), ],
                      na.rm = TRUE)
print(m3_1v3)
```

Wilcoxon rank sum test with continuity correction

```
data: p_AltBaselineMTM by altair.resultPatW
W = 449, p-value = 0.09739
alternative hypothesis: true location shift is not equal to 0
```

[Hide](#)

```
m3_2v3 <- wilcox.test(p_AltBaselineMTM ~ altair.resultPatW,
                      data = circ_data[circ_data$altair.resultPatW %in% c("Transient clearance", "No clearance"), ],
                      na.rm = TRUE)
print(m3_2v3)
```

Wilcoxon rank sum test with continuity correction

```
data: p_AltBaselineMTM by altair.resultPatW
W = 2201, p-value = 5.386e-06
alternative hypothesis: true location shift is not equal to 0
```

[Hide](#)

```
# Create a table with p-values
p_value_table <- data.frame(
  Comparison = c("Sustained vs Transient", "Sustained vs No Clearance", "Transient vs No Clearance"),
  P_Value = c(m3_1v2$p.value, m3_1v3$p.value, m3_2v3$p.value)
)
print(p_value_table)
```

|   | Comparison                | P_Value      |
|---|---------------------------|--------------|
| 1 | Sustained vs Transient    | 7.395008e-01 |
| 2 | Sustained vs No Clearance | 9.738952e-02 |
| 3 | Transient vs No Clearance | 5.386244e-06 |

#DFS1 by ctDNA Clearance with 3 groups - All stages

[Hide](#)

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$altair.group.SAP_MSv2!="1b = Exclude: No on-Tx TPs",]

circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("No clearance", "Transient clearance", "Sustained clearance"))
survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.resultPatW, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
~
      altair.resultPatW, data = circ_data)
```

|                                       | n   | events | median | 0.95LCL | 0.95UCL |
|---------------------------------------|-----|--------|--------|---------|---------|
| altair.resultPatW=No clearance        | 147 | 140    | 4.37   | 3.75    | 5.82    |
| altair.resultPatW=Transient clearance | 63  | 45     | 11.76  | 11.01   | 15.54   |
| altair.resultPatW=Sustained clearance | 25  | 5      | NA     | 33.91   | NA      |

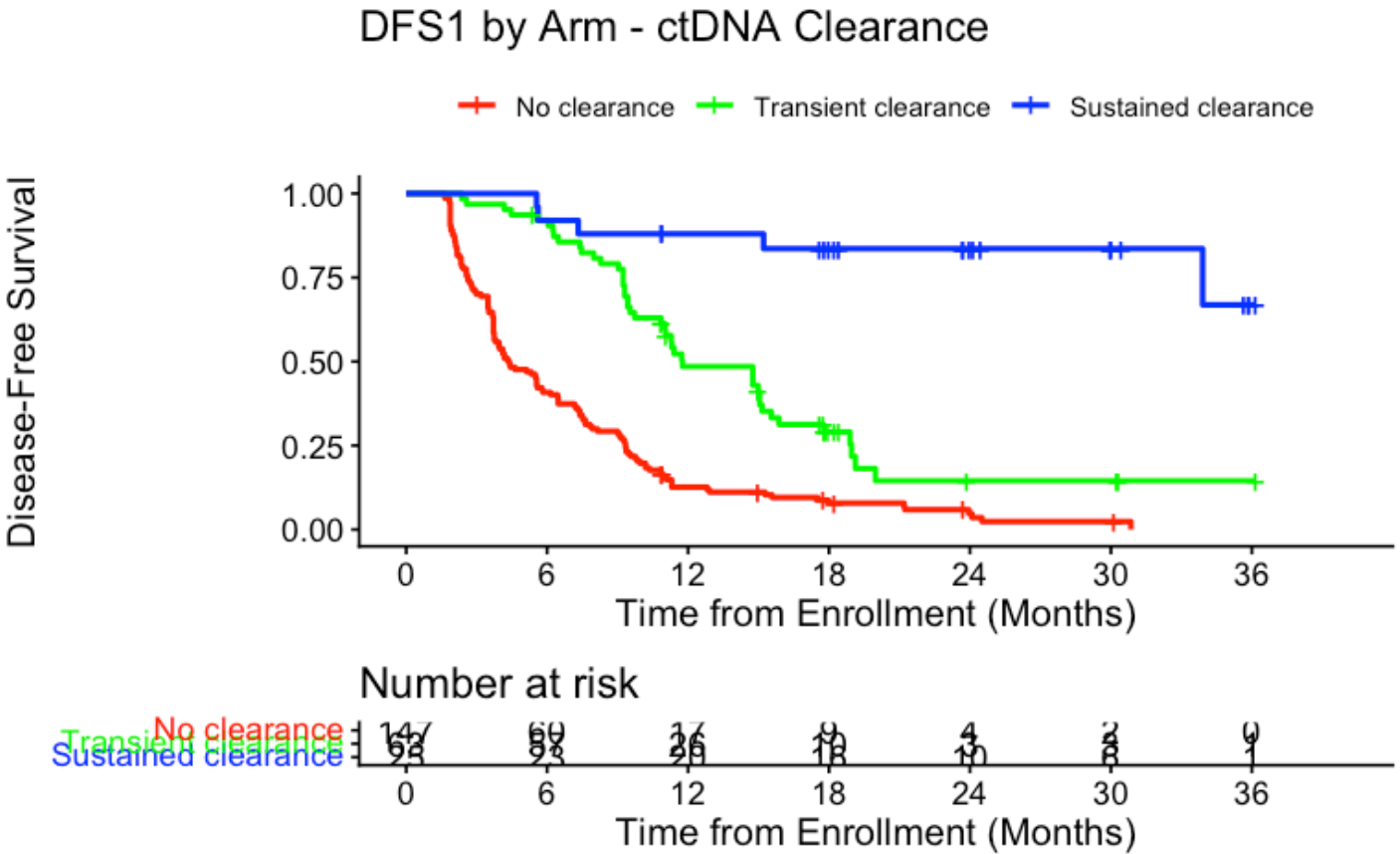
Hide

```
event_summary <- circ_data %>%
  group_by(altair.resultPatW) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 3 × 5
  altair.resultPatW Total Events Fraction Percentage
<fct>             <int> <int>    <dbl>    <dbl>
1 No clearance      147   140    0.952    95.2
2 Transient clearance 63    45    0.714    71.4
3 Sustained clearance 25     5    0.2      20
```

Hide

```
surv_object <- Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.resultPatW, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","green","blue"), title="DFS1 by Arm - ctDNA Clearance", ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("No clearance", "Transient clearance", "Sustained clearance"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```



```
Call: survfit(formula = surv_object ~ altair.resultPatW, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

```
altair.resultPatW=No clearance
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 60     | 87      | 0.4082   | 0.0405  | 0.3284       | 0.4862       |
| 12   | 17     | 41      | 0.1262   | 0.0277  | 0.0782       | 0.1861       |
| 18   | 9      | 6       | 0.0787   | 0.0232  | 0.0412       | 0.1319       |
| 24   | 4      | 3       | 0.0472   | 0.0199  | 0.0183       | 0.0975       |

```
altair.resultPatW=Transient clearance
```

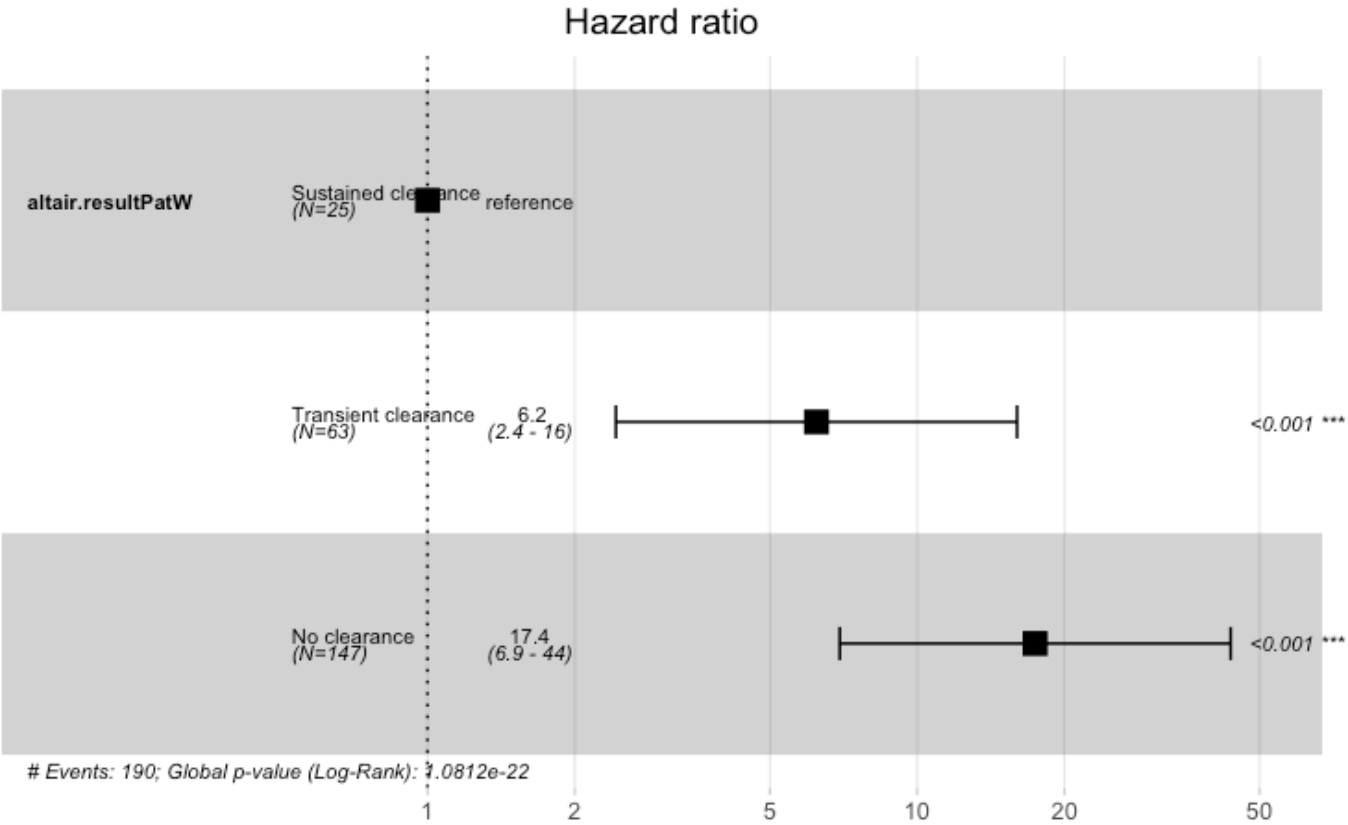
| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 57     | 5       | 0.920    | 0.0342  | 0.8192       | 0.966        |
| 12   | 26     | 26      | 0.485    | 0.0653  | 0.3530       | 0.605        |
| 18   | 10     | 10      | 0.290    | 0.0619  | 0.1767       | 0.414        |
| 24   | 3      | 4       | 0.145    | 0.0600  | 0.0531       | 0.281        |

```
altair.resultPatW=Sustained clearance
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 23     | 2       | 0.920    | 0.0543  | 0.716        | 0.979        |
| 12   | 20     | 1       | 0.880    | 0.0650  | 0.673        | 0.960        |
| 18   | 16     | 1       | 0.836    | 0.0752  | 0.619        | 0.935        |
| 24   | 10     | 0       | 0.836    | 0.0752  | 0.619        | 0.935        |

[Hide](#)

```
circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("Sustained c
learance", "Transient clearance", "No clearance"))
cox_fit <- coxph(surv_object ~ altair.resultPatW, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ altair.resultPatW, data = circ\_data)

n= 235, number of events= 190

|                                      | coef   | exp(coef) | se(coef) | z     | Pr(> z ) |     |
|--------------------------------------|--------|-----------|----------|-------|----------|-----|
| altair.resultPatWTransient clearance | 1.8292 | 6.2286    | 0.4812   | 3.801 | 0.000144 | *** |
| altair.resultPatWNo clearance        | 2.8576 | 17.4200   | 0.4690   | 6.094 | 1.1e-09  | *** |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|--------------------------------------|-----------|------------|-----------|-----------|
| altair.resultPatWTransient clearance | 6.229     | 0.16055    | 2.425     | 16.00     |
| altair.resultPatWNo clearance        | 17.420    | 0.05741    | 6.948     | 43.67     |

Concordance= 0.686 (se = 0.017 )  
Likelihood ratio test= 101.2 on 2 df, p=<2e-16  
Wald test = 64.59 on 2 df, p=9e-15  
Score (logrank) test = 87.72 on 2 df, p=<2e-16

Hide

```
cox_fit_summary <- summary(cox_fit)
```

## #OS by ctDNA Clearance with 3 groups - All stages

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data <- circ_data[circ_data$altair.group.SAP_MSV2!="1b = Exclude: No on-Tx TPs",]

circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("No clearance", "Transient clearance", "Sustained clearance"))
survfit(Surv(time = circ_data$OS.months, event = circ_data$p_evt0S)~altair.resultPatW, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$OS.months, event = circ\_data\$p\_evt0S) ~ altair.resultPatW, data = circ\_data)

|                                       | n   | events | median | 0.95LCL | 0.95UCL |
|---------------------------------------|-----|--------|--------|---------|---------|
| altair.resultPatW=No clearance        | 147 | 19     | NA     | 40.7    | NA      |
| altair.resultPatW=Transient clearance | 63  | 2      | NA     | NA      | NA      |
| altair.resultPatW=Sustained clearance | 25  | 0      | NA     | NA      | NA      |

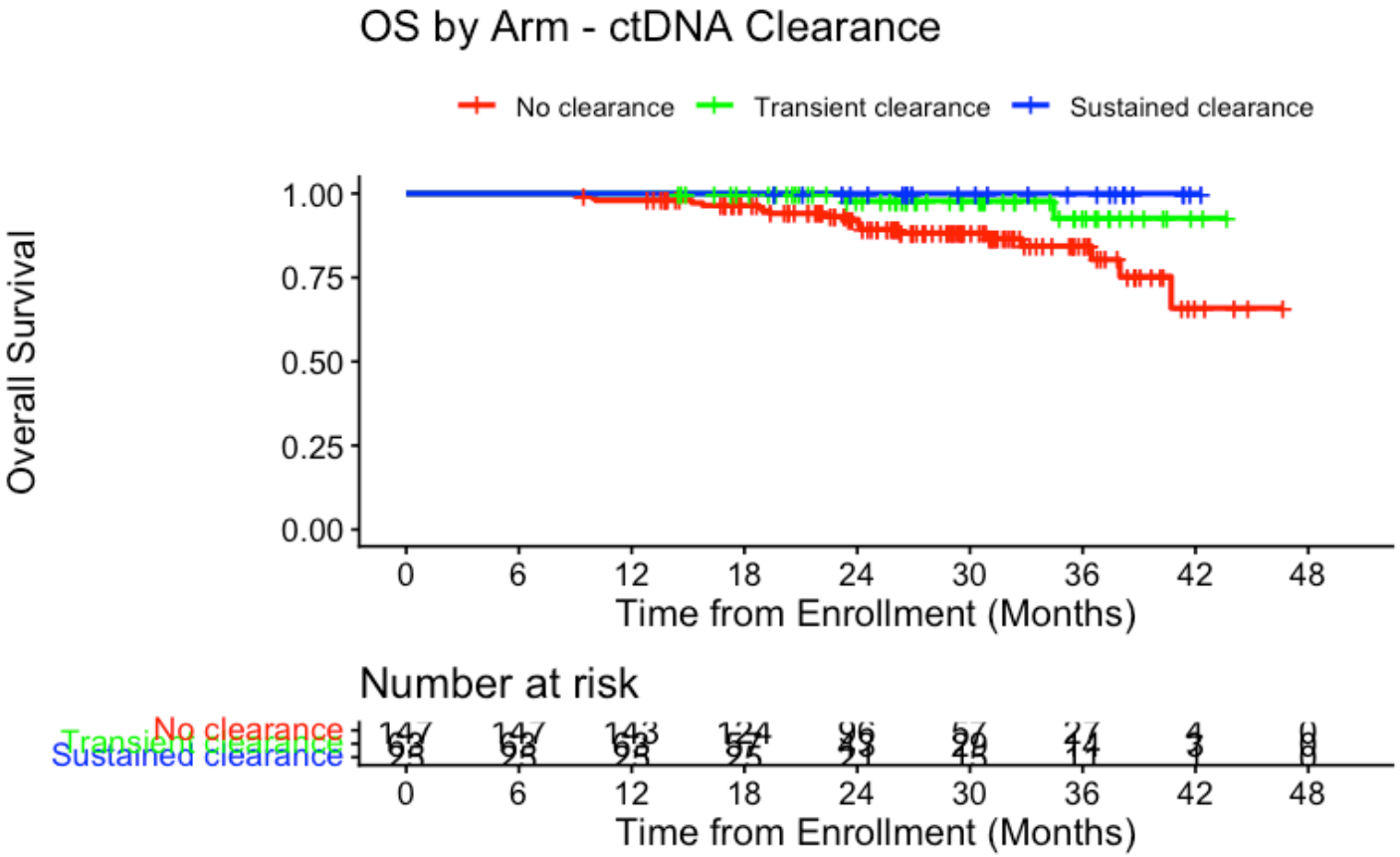
Hide

```
event_summary <- circ_data %>%
  group_by(altair.resultPatW) %>%
  summarise(
    Total = n(),
    Events = sum(p_evt0S),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 3 × 5
  altair.resultPatW Total Events Fraction Percentage
  <fct>           <int> <int>   <dbl>      <dbl>
1 No clearance    147    19  0.129      12.9
2 Transient clearance 63     2  0.0317     3.17
3 Sustained clearance 25     0  0          0
```

Hide

```
surv_object <- Surv(time = circ_data$OS.months, event = circ_data$p_evt0S)
KM_curve <- survfit(surv_object ~ altair.resultPatW, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","green","blue"), title="OS by Arm - ctDNA Clearance", ylab= "Overall Survival", xlab="Time from Enrollment (Months)", legend.labs=c("No clearance", "Transient clearance", "Sustained clearance"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.resultPatW, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

```
altair.resultPatW=No clearance
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 147    | 0       | 1.000    | 0.0000  | 1.000        | 1.000        |
| 12   | 143    | 3       | 0.979    | 0.0117  | 0.938        | 0.993        |
| 18   | 124    | 2       | 0.965    | 0.0155  | 0.917        | 0.985        |
| 24   | 96     | 5       | 0.923    | 0.0237  | 0.860        | 0.958        |

```
altair.resultPatW=Transient clearance
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 63     | 0       | 1.000    | 0.0000  | 1.000        | 1.000        |
| 12   | 63     | 0       | 1.000    | 0.0000  | 1.000        | 1.000        |
| 18   | 57     | 0       | 1.000    | 0.0000  | 1.000        | 1.000        |
| 24   | 43     | 1       | 0.978    | 0.0215  | 0.856        | 0.997        |

```
altair.resultPatW=Sustained clearance
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 6    | 25     | 0       | 1        | 0       | 1            | 1            |
| 12   | 25     | 0       | 1        | 0       | 1            | 1            |
| 18   | 25     | 0       | 1        | 0       | 1            | 1            |
| 24   | 21     | 0       | 1        | 0       | 1            | 1            |

[Hide](#)

```
circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("Sustained c
learance", "Transient clearance", "No clearance"))
cox_fit <- coxphf(surv_object ~ altair.resultPatW, data=circ_data)
summary(cox_fit)
```

```
coxphf(formula = surv_object ~ altair.resultPatW, data = circ_data)
```

Model fitted by Penalized ML

Confidence intervals and p-values by Profile Likelihood

|                                      |            | coef     | se(coef) | exp(coef) | lower 0.95 | upper 0.95 |
|--------------------------------------|------------|----------|----------|-----------|------------|------------|
| Chisq                                | p          |          |          |           |            |            |
| altair.resultPatWTransient clearance |            | 1.004483 | 1.609325 | 2.730495  | 0.221318   | 377.0473   |
| 0.5145106                            | 0.47319272 |          |          |           |            |            |
| altair.resultPatWNo clearance        |            | 2.324182 | 1.489576 | 10.218322 | 1.391342   | 1302.7825  |
| 5.8730349                            | 0.01537452 |          |          |           |            |            |

Likelihood ratio test=9.503886 on 2 df, p=0.008634899, n=235

Wald test = 5.724023 on 2 df, p = 0.05715369

Covariance-Matrix:

|                                      | altair.resultPatWTransient clearance | altair.resultP |
|--------------------------------------|--------------------------------------|----------------|
| atWNo clearance                      |                                      |                |
| altair.resultPatWTransient clearance | 2.589926                             |                |
| 2.162201                             |                                      |                |
| altair.resultPatWNo clearance        | 2.162201                             |                |
| 2.218837                             |                                      |                |

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

```
coxphf(formula = surv_object ~ altair.resultPatW, data = circ_data)
```

Model fitted by Penalized ML

Confidence intervals and p-values by Profile Likelihood

|                                      |            | coef     | se(coef) | exp(coef) | lower 0.95 | upper 0.95 |
|--------------------------------------|------------|----------|----------|-----------|------------|------------|
| Chisq                                | p          |          |          |           |            |            |
| altair.resultPatWTransient clearance |            | 1.004483 | 1.609325 | 2.730495  | 0.221318   | 377.0473   |
| 0.5145106                            | 0.47319272 |          |          |           |            |            |
| altair.resultPatWNo clearance        |            | 2.324182 | 1.489576 | 10.218322 | 1.391342   | 1302.7825  |
| 5.8730349                            | 0.01537452 |          |          |           |            |            |

Likelihood ratio test=9.503886 on 2 df, p=0.008634899, n=235

Wald test = 5.724023 on 2 df, p = 0.05715369

Covariance-Matrix:

|                                      | altair.resultPatWTransient clearance | altair.resultP |
|--------------------------------------|--------------------------------------|----------------|
| atWNo clearance                      |                                      |                |
| altair.resultPatWTransient clearance | 2.589926                             |                |
| 2.162201                             |                                      |                |
| altair.resultPatWNo clearance        | 2.162201                             |                |
| 2.218837                             |                                      |                |

#DFS1 by ctDNA Clearance with 3 groups TAS-102 arm - All stages

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels = c("Control", "Experimental"), labels = c("Placebo", "FTD/TPI"))
circ_data <- circ_data[circ_data$altair.Arm=="FTD/TPI",]
circ_data <- circ_data[circ_data$altair.group.SAP_MSv2!="1b = Exclude: No on-Tx TPs",]

circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("No clearance", "Transient clearance", "Sustained clearance"))
survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.resultPatW, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$DFS.months, event = circ\_data\$p\_evtDFS1b) ~ altair.resultPatW, data = circ\_data)

|                                       | n  | events | median | 0.95LCL | 0.95UCL |
|---------------------------------------|----|--------|--------|---------|---------|
| altair.resultPatW=No clearance        | 62 | 60     | 5.67   | 3.94    | 7.82    |
| altair.resultPatW=Transient clearance | 47 | 35     | 11.30  | 9.73    | 15.15   |
| altair.resultPatW=Sustained clearance | 11 | 2      | NA     | NA      | NA      |

Hide

```
event_summary <- circ_data %>%
  group_by(altair.resultPatW) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

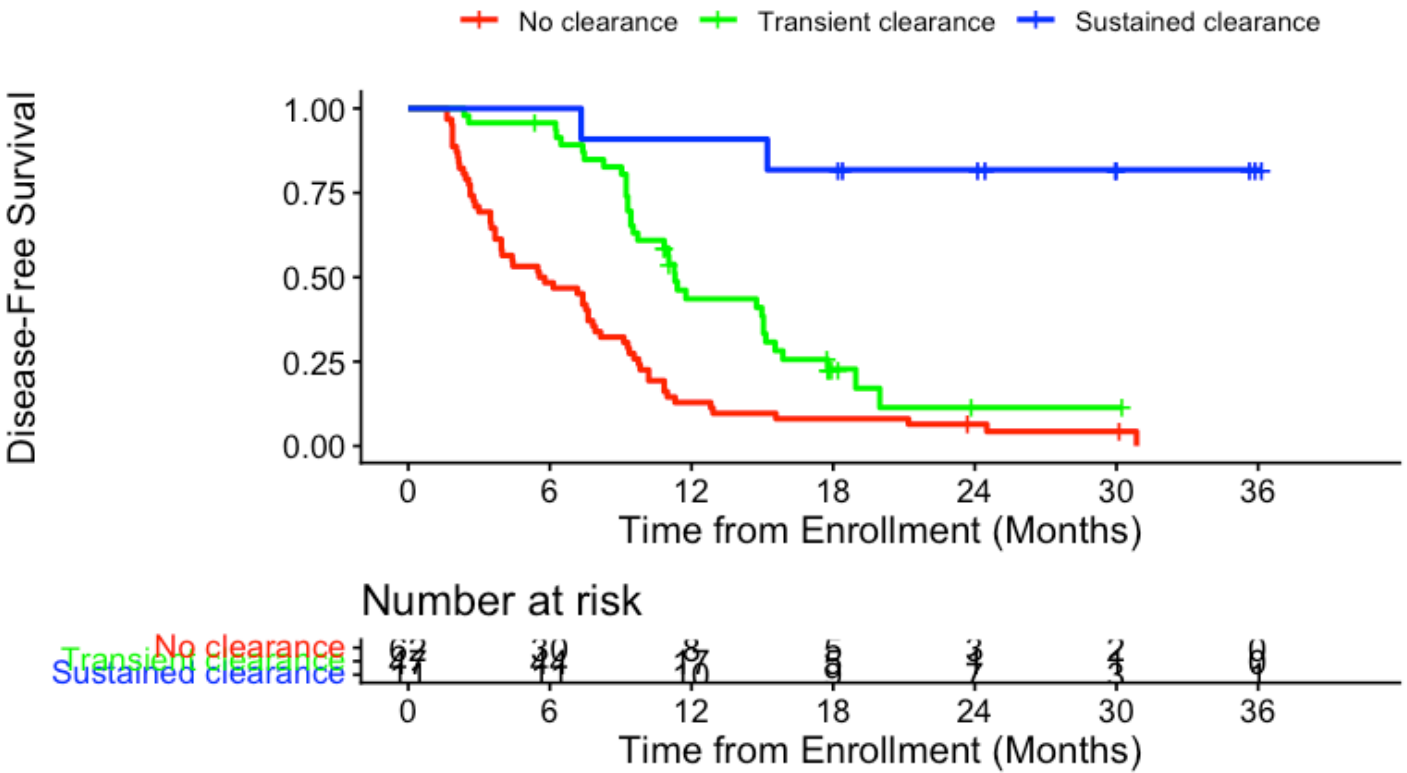
```
# A tibble: 3 × 5
  altair.resultPatW Total Events Fraction Percentage
  <fct>           <int>  <int>    <dbl>      <dbl>
1 No clearance      62    60    0.968      96.8
2 Transient clearance 47    35    0.745      74.5
3 Sustained clearance 11     2    0.182      18.2
```

Hide

```
surv_object <-Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.resultPatW, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","green","blue"), title="DFS1 by Arm - ctDNA Clearance", ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("No clearance", "Transient clearance", "Sustained clearance"), legend.title="")
```



DFS1 by Arm - ctDNA Clearance



Hide

```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.resultPatW, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

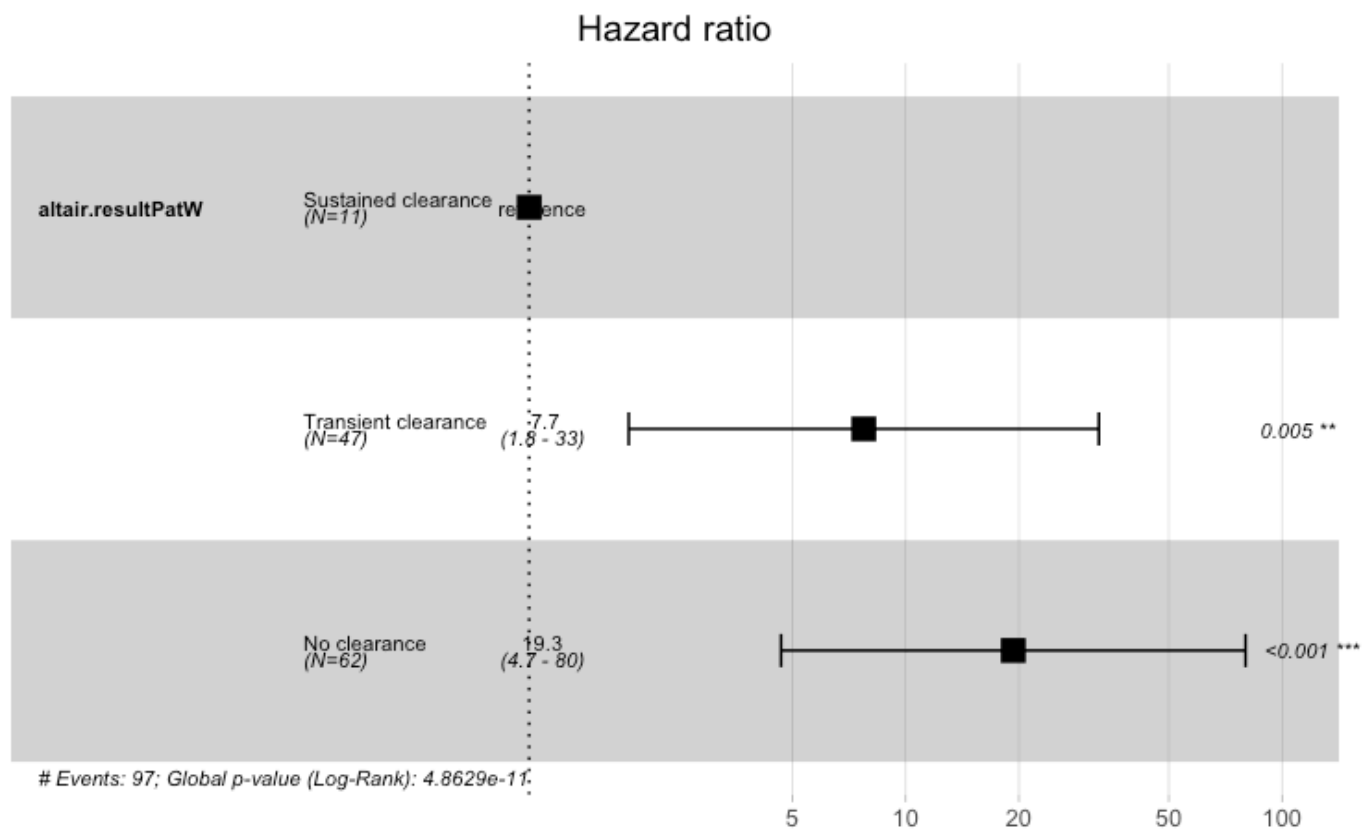
|                                |        |         |          |         |              |              |  |
|--------------------------------|--------|---------|----------|---------|--------------|--------------|--|
| altair.resultPatW=No clearance |        |         |          |         |              |              |  |
| time                           | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                              | 30     | 32      | 0.4839   | 0.0635  | 0.3554       | 0.601        |  |
| 12                             | 8      | 22      | 0.1290   | 0.0426  | 0.0603       | 0.225        |  |
| 18                             | 5      | 3       | 0.0806   | 0.0346  | 0.0297       | 0.165        |  |
| 24                             | 3      | 1       | 0.0645   | 0.0312  | 0.0208       | 0.144        |  |

|                                       |        |         |          |         |              |              |  |
|---------------------------------------|--------|---------|----------|---------|--------------|--------------|--|
| altair.resultPatW=Transient clearance |        |         |          |         |              |              |  |
| time                                  | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                                     | 44     | 2       | 0.957    | 0.0294  | 0.8404       | 0.989        |  |
| 12                                    | 17     | 23      | 0.436    | 0.0758  | 0.2860       | 0.577        |  |
| 18                                    | 5      | 8       | 0.228    | 0.0666  | 0.1133       | 0.366        |  |
| 24                                    | 1      | 2       | 0.114    | 0.0660  | 0.0257       | 0.276        |  |

|                                       |        |         |          |         |              |              |  |
|---------------------------------------|--------|---------|----------|---------|--------------|--------------|--|
| altair.resultPatW=Sustained clearance |        |         |          |         |              |              |  |
| time                                  | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                                     | 11     | 0       | 1.000    | 0.0000  | 1.000        | 1.000        |  |
| 12                                    | 10     | 1       | 0.909    | 0.0867  | 0.508        | 0.987        |  |
| 18                                    | 9      | 1       | 0.818    | 0.1163  | 0.447        | 0.951        |  |
| 24                                    | 7      | 0       | 0.818    | 0.1163  | 0.447        | 0.951        |  |

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```
circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("Sustained c  
clearance", "Transient clearance", "No clearance"))  
cox_fit <- coxph(surv_object ~ altair.resultPatW, data=circ_data)  
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ altair.resultPatW, data = circ_data)
```

n= 120, number of events= 97

|                                      | coef   | exp(coef) | se(coef) | z     | Pr(> z ) |     |
|--------------------------------------|--------|-----------|----------|-------|----------|-----|
| altair.resultPatWTransient clearance | 2.0466 | 7.7414    | 0.7334   | 2.791 | 0.00526  | **  |
| altair.resultPatWNo clearance        | 2.9611 | 19.3193   | 0.7247   | 4.086 | 4.39e-05 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|--------------------------------------|-----------|------------|-----------|-----------|
| altair.resultPatWTransient clearance | 7.741     | 0.12918    | 1.839     | 32.59     |
| altair.resultPatWNo clearance        | 19.319    | 0.05176    | 4.668     | 79.96     |

Concordance= 0.693 (se = 0.024 )

Likelihood ratio test= 47.49 on 2 df, p=5e-11

Wald test = 31.05 on 2 df, p=2e-07

Score (logrank) test = 42.54 on 2 df, p=6e-10

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

#DFS1 by ctDNA Clearance with 3 groups Placebo arm - All stages

[Hide](#)

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20240729 Dataset.csv")
circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels = c("Control", "Experimental"), labels = c("Placebo", "FTD/TPI"))
circ_data <- circ_data[circ_data$altair.Arm=="Placebo",]
circ_data <- circ_data[circ_data$altair.group.SAP_MSv2!="1b = Exclude: No on-Tx TPs",]

circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("No clearance", "Transient clearance", "Sustained clearance"))
survfit(Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)~altair.resultPatW, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b) ~
altair.resultPatW, data = circ_data)
```

|                                       | n  | events | median | 0.95LCL | 0.95UCL |
|---------------------------------------|----|--------|--------|---------|---------|
| altair.resultPatW=No clearance        | 85 | 80     | 4.11   | 3.71    | 5.55    |
| altair.resultPatW=Transient clearance | 16 | 10     | 14.75  | 7.98    | NA      |
| altair.resultPatW=Sustained clearance | 14 | 3      | 33.91  | 33.91   | NA      |

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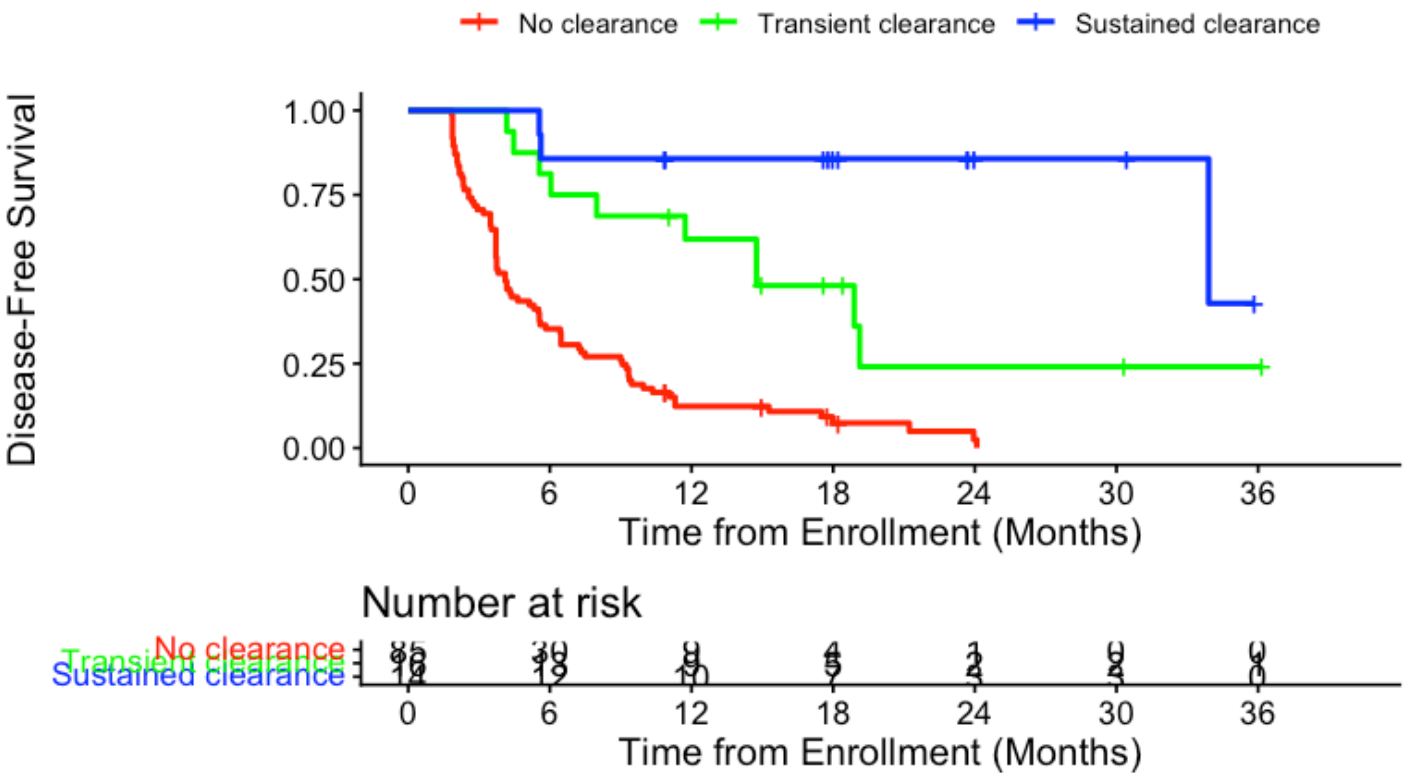
```
event_summary <- circ_data %>%
  group_by(altair.resultPatW) %>%
  summarise(
    Total = n(),
    Events = sum(p_evtDFS1b),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

```
# A tibble: 3 × 5
  altair.resultPatW   Total Events Fraction Percentage
  <fct>             <int>   <int>   <dbl>     <dbl>
1 No clearance       85     80    0.941     94.1
2 Transient clearance 16     10    0.625     62.5
3 Sustained clearance 14      3    0.214     21.4
```

[Hide](#)

```
surv_object <- Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.resultPatW, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("red","green","blue"), title="DFS1 by Arm - ctDNA Clearance", ylab= "Disease-Free Survival", xlab="Time from Enrollment (Months)", legend.labs=c("No clearance", "Transient clearance", "Sustained clearance"), legend.title="")
```

DFS1 by Arm - ctDNA Clearance



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```
summary(KM_curve, times= c(6, 12, 18, 24))
```

```
Call: survfit(formula = surv_object ~ altair.resultPatW, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

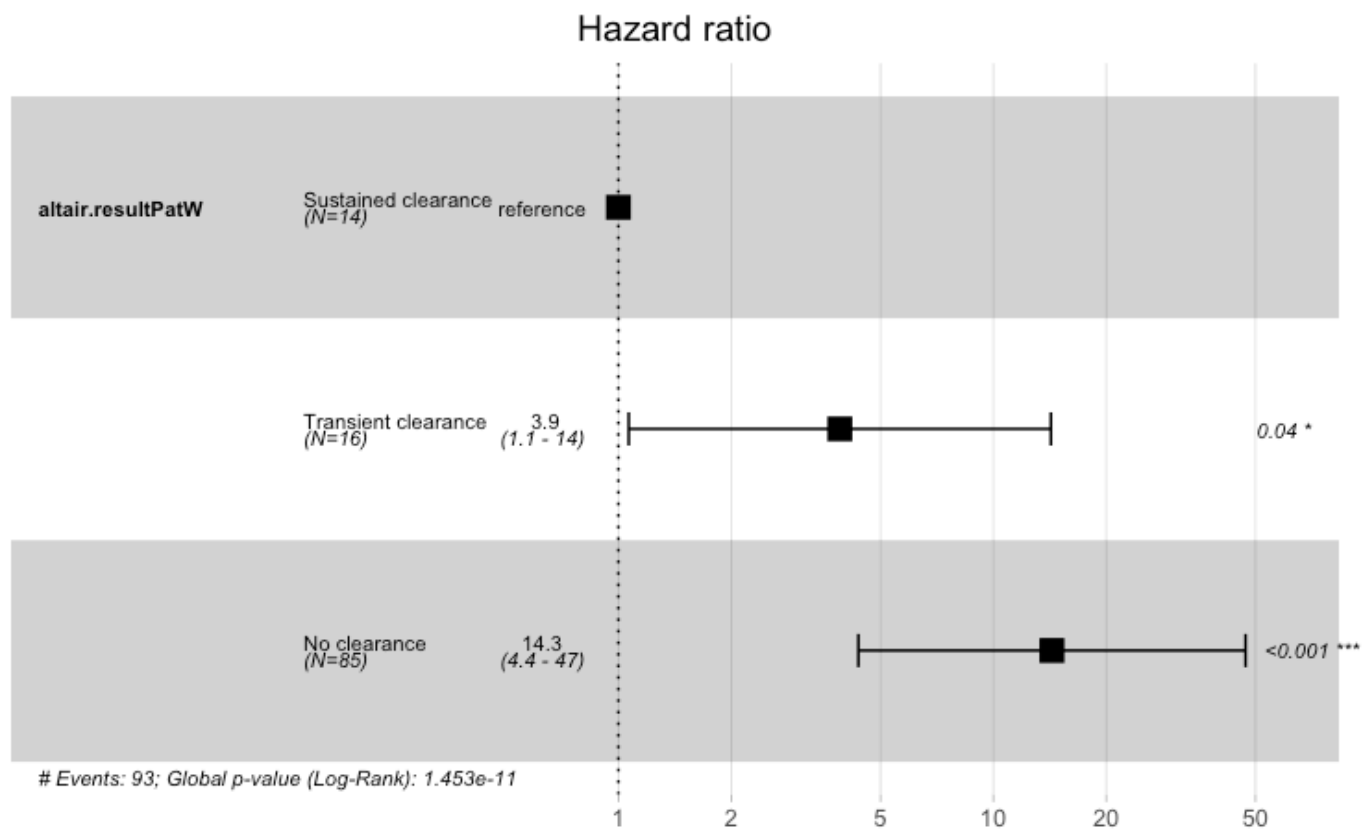
| altair.resultPatW=No clearance |        |         |          |         |              |              |  |
|--------------------------------|--------|---------|----------|---------|--------------|--------------|--|
| time                           | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                              | 30     | 55      | 0.3529   | 0.0518  | 0.25334      | 0.454        |  |
| 12                             | 9      | 19      | 0.1235   | 0.0365  | 0.06335      | 0.205        |  |
| 18                             | 4      | 3       | 0.0741   | 0.0314  | 0.02790      | 0.151        |  |
| 24                             | 1      | 2       | 0.0247   | 0.0227  | 0.00242      | 0.103        |  |

| altair.resultPatW=Transient clearance |        |         |          |         |              |              |  |
|---------------------------------------|--------|---------|----------|---------|--------------|--------------|--|
| time                                  | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                                     | 13     | 3       | 0.812    | 0.0976  | 0.5246       | 0.935        |  |
| 12                                    | 9      | 3       | 0.619    | 0.1230  | 0.3393       | 0.808        |  |
| 18                                    | 5      | 2       | 0.481    | 0.1285  | 0.2241       | 0.699        |  |
| 24                                    | 2      | 2       | 0.241    | 0.1364  | 0.0447       | 0.520        |  |

| altair.resultPatW=Sustained clearance |        |         |          |         |              |              |  |
|---------------------------------------|--------|---------|----------|---------|--------------|--------------|--|
| time                                  | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                                     | 12     | 2       | 0.857    | 0.0935  | 0.539        | 0.962        |  |
| 12                                    | 10     | 0       | 0.857    | 0.0935  | 0.539        | 0.962        |  |
| 18                                    | 7      | 0       | 0.857    | 0.0935  | 0.539        | 0.962        |  |
| 24                                    | 3      | 0       | 0.857    | 0.0935  | 0.539        | 0.962        |  |

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```
circ_data$altair.resultPatW <- factor(circ_data$altair.resultPatW, levels=c("Sustained c  
learance", "Transient clearance", "No clearance"))  
cox_fit <- coxph(surv_object ~ altair.resultPatW, data=circ_data)  
ggforest(cox_fit, data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ altair.resultPatW, data = circ_data)
```

n= 115, number of events= 93

|                                      | coef   | exp(coef) | se(coef) | z     | Pr(> z )     |
|--------------------------------------|--------|-----------|----------|-------|--------------|
| altair.resultPatWTransient clearance | 1.3595 | 3.8944    | 0.6613   | 2.056 | 0.0398 *     |
| altair.resultPatWNo clearance        | 2.6623 | 14.3296   | 0.6066   | 4.389 | 1.14e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|--------------------------------------|-----------|------------|-----------|-----------|
| altair.resultPatWTransient clearance | 3.894     | 0.25678    | 1.066     | 14.23     |
| altair.resultPatWNo clearance        | 14.330    | 0.06979    | 4.364     | 47.05     |

Concordance= 0.659 (se = 0.024 )

Likelihood ratio test= 49.91 on 2 df, p=1e-11

Wald test = 29.7 on 2 df, p=4e-07

Score (logrank) test = 40.71 on 2 df, p=1e-09

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

#Central Review Imaging analysis #DFS1 by TAS vs Placebo - Central review data - All stages & stratified for Stage & ctDNA 1mo post-surgery

[Hide](#)

```
rm(list = ls())
setwd("~/Downloads")
circ_data <- read.csv("Altair 20250903 Central Imaging Dataset.csv")
circ_data <- as.data.frame(circ_data)

circ_data$altair.Arm <- factor(circ_data$altair.Arm, levels = c("Control","Experimental"), labels = c("Placebo","FTD/TPI"))
circ_data$Disease.Stage <- factor(circ_data$Disease.Stage, levels = c("Stage II or lower","StageIII","M1"))
circ_data$ctDNA1mo <- factor(circ_data$ctDNA1mo, levels = c("NEGATIVE","POSITIVE"), labels = c("Negative","Positive"))

event_summary <- circ_data %>% group_by(altair.Arm) %>% summarise(Total = n(), Events = sum(p_evtDFS1b), Fraction = Events / n(), Percentage = (Events / n()) * 100)
print(event_summary)
```

# A tibble: 2 × 5

|   | altair.Arm | Total | Events | Fraction | Percentage |
|---|------------|-------|--------|----------|------------|
|   | <fct>      | <int> | <int>  | <dbl>    | <dbl>      |
| 1 | Placebo    | 121   | 100    | 0.826    | 82.6       |
| 2 | FTD/TPI    | 122   | 96     | 0.787    | 78.7       |

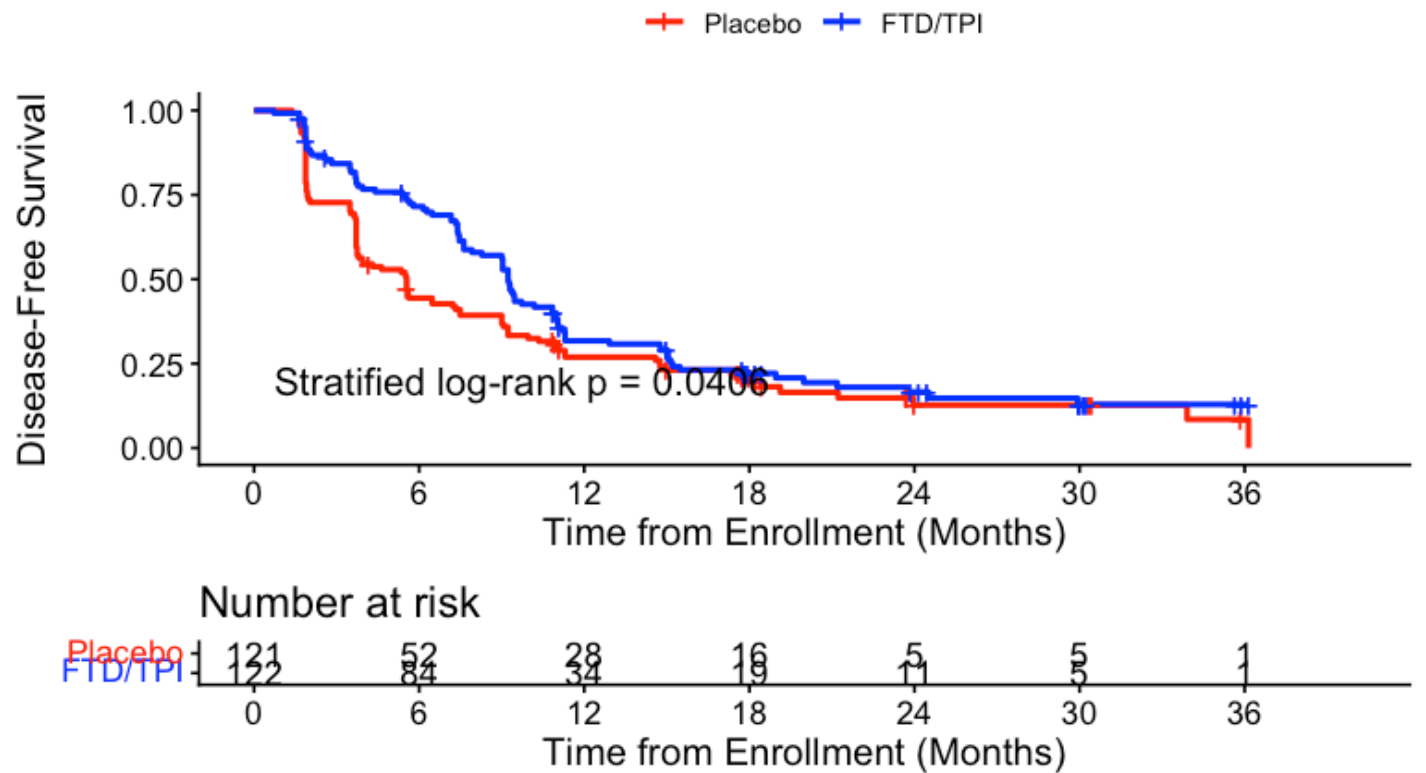
```
surv_object <- Surv(time = circ_data$DFS.months, event = circ_data$p_evtDFS1b)
KM_curve <- survfit(surv_object ~ altair.Arm, data = circ_data, conf.int = 0.95, conf.type = "log-log")

# --- Stratified log-rank test (exactly as in primary analysis) ---
# Use survdiff with the same stratification factors
sd_strat <- survdiff(surv_object ~ altair.Arm + strata(Disease.Stage, ctDNA1mo), data = circ_data)
# For two groups, df = 1
p_strat_logrank <- 1 - pchisq(sd_strat$chisq, df = 1)

# KM plot with the stratified p-value displayed
ggsurvplot(KM_curve, data = circ_data, pval = sprintf("Stratified log-rank p = %.4f", p_strat_logrank), conf.int = FALSE, risk.table = TRUE, break.time.by = 6, palette = c("red", "blue"), title = "DFS1 by Arm - All Patients", ylab = "Disease-Free Survival", xlab = "Time from Enrollment (Months)", legend.labs = c("Placebo", "FTD/TPI"), legend.title = "")
```



DFS1 by Arm - All Patients



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```
print(summary(KM_curve, times = c(6, 12, 18, 24)))
```

Call: `survfit(formula = surv_object ~ altair.Arm, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

| altair.Arm=Placebo |        |         |          |         |              |              |  |
|--------------------|--------|---------|----------|---------|--------------|--------------|--|
| time               | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                  | 52     | 67      | 0.444    | 0.0453  | 0.3542       | 0.530        |  |
| 12                 | 28     | 20      | 0.269    | 0.0412  | 0.1923       | 0.352        |  |
| 18                 | 16     | 6       | 0.210    | 0.0386  | 0.1397       | 0.290        |  |
| 24                 | 5      | 5       | 0.127    | 0.0379  | 0.0649       | 0.212        |  |

| altair.Arm=FTD/TPI |        |         |          |         |              |              |  |
|--------------------|--------|---------|----------|---------|--------------|--------------|--|
| time               | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 6                  | 84     | 34      | 0.716    | 0.0413  | 0.6254       | 0.788        |  |
| 12                 | 34     | 46      | 0.318    | 0.0434  | 0.2347       | 0.403        |  |
| 18                 | 19     | 10      | 0.222    | 0.0395  | 0.1494       | 0.303        |  |
| 24                 | 11     | 4       | 0.166    | 0.0381  | 0.0996       | 0.247        |  |

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```
cox_fit_stratified <- coxph(surv_object ~ altair.Arm + strata(Disease.Stage) + strata(ct
DNA1mo), data = circ_data)
cox_fit_summary_stratified <- summary(cox_fit_stratified)
print(cox_fit_summary_stratified)
```

Call:

```
coxph(formula = surv_object ~ altair.Arm + strata(Disease.Stage) +
      strata(ctDNA1mo), data = circ_data)
```

n= 243, number of events= 196

|                   | coef    | exp(coef) | se(coef) | z      | Pr(> z ) |
|-------------------|---------|-----------|----------|--------|----------|
| altair.ArmFTD/TPI | -0.3048 | 0.7372    | 0.1462   | -2.085 | 0.0371 * |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| altair.ArmFTD/TPI | 0.7372    | 1.356      | 0.5536    | 0.9819    |

Concordance= 0.565 (se = 0.021 )

Likelihood ratio test= 4.34 on 1 df, p=0.04

Wald test = 4.35 on 1 df, p=0.04

Score (logrank) test = 4.38 on 1 df, p=0.04

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```
HR_stratified      <- cox_fit_summary_stratified$conf.int["altair.ArmFTD/TPI","exp(coe
f)"]
lower_CI_stratified <- cox_fit_summary_stratified$conf.int["altair.ArmFTD/TPI","lower .9
5"]
upper_CI_stratified <- cox_fit_summary_stratified$conf.int["altair.ArmFTD/TPI","upper .9
5"]
```

```
# --- Use SCORE (log-rank) p-value from the Cox fit ---
if (!is.null(cox_fit_summary_stratified$score)) {
  score_stat <- cox_fit_summary_stratified$score
} else {
  score_stat <- as.numeric(cox_fit_summary_stratified$sctest["test"])
}
p_value_score <- 1 - pchisq(score_stat, df = 1)
```

```
# (Optional) Wald p-value for reference
p_value_wald <- as.numeric(cox_fit_summary_stratified$wald["pvalue"])
```

```
# Label using SCORE p-value
label_text_stratified <- paste0(
  "HR = ", round(HR_stratified, 2),
  " (", round(lower_CI_stratified, 2), "-",
  round(upper_CI_stratified, 2), "); p (score) = ",
  format.pval(p_value_score, digits = 3)
)
print(label_text_stratified)
```

```
[1] "HR = 0.74 (0.55-0.98); p (score) = 0.036"
```

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```
#print both p-values side-by-side:  
cat(sprintf("Stratified log-rank (survdiff) p = %.4f\n", p_strat_logrank))
```

```
Stratified log-rank (survdiff) p = 0.0406
```

[Hide](#)

```
cat(sprintf("Cox score-test p = %.4f (Wald p = %.4f)\n", p_value_score, p_value_wald))
```

```
Cox score-test p = 0.0364 (Wald p = 0.0371)
```